Q/R/M/C: A Stage-Decomposition Measurement Framework for Memory-Based Navigation

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Abstract

Setting. Success-only metrics collapse qualitatively different memory-based navigation failures into a single scalar, hiding *which stage* of the retrieval pipeline is responsible. This paper contributes an *evaluation protocol*: the Q/R/M/C stage decomposition (**Q**uery formation, **R**etrieval satisfaction, target **M**aterialization, **C**ompletion after retrieval), operationalised so that the four conditional rates are computable directly from standard trajectory logs (Definition 1). The arithmetic is the chain rule, deliberately so — just as decomposing accuracy into precision and recall added no mathematics but changed what the field could *see*, the value here is identifiability from logs and what the decomposition reveals, not a theorem. **What it decomposes** is important to state up front: the product of the four rates converges to *semantic-path completion* $C^{\star}$, not to raw task success Y; because an episode can succeed without traversing the retrieval path, Y can exceed $C^{\star}$, so Q/R/M/C localises *where the semantic route fails*, not every route to success. **Single-agent measurement.** On a 10-seed MiniGrid benchmark (FourRooms, LavaGapS7) and a BabyAI portability check, the framework localises bottlenecks: hazard recovery is retrieval-limited (oracle retrieval lifts success $0.39 \to 0.60$), goal-region rejoin is downstream-limited. Log inspection refines the attribution: candidate ranking is near-optimal (96.9%) while candidate generation fails at 91% of decision points — a memory-accumulation, not a ranking, limit. **Multi-agent measurement.** On a 35,640-run multi-agent sweep over four communication architectures (independent, shared-bus, central-aggregator, peer-broadcast at eight cadences $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$, 40 seeds), the bottleneck-shift between the M and C links appears directly as a negative within-cadence correlation peaking at $K=4$ (Pearson -0.61); mean time-to-success has an interior minimum at $K=8$ (robust in location but shallow in margin — not resolved from the adjacent fast cadences). Both slope signs (Empirical Claim 1) hold as moderate, cluster-robust effect sizes (Spearman -0.53 and $+0.43$ vs. K, 95% CIs $[-0.59, -0.47]$ and $[+0.37, +0.50]$ over 81 configuration cells; Appendix F.3). Non-claims (notably on the conditional product Φ) are consolidated in one place (§1, Scope and non-claims). **Diagnostic use.** The same four rates also expose why learned policies with implicit communication can be competitive on success while collapsing Q/R/M/C observability: their communication vectors are never informatively empty and they lack an explicit target-lock interface. We scope this claim explicitly: Q/R/M/C is separately observable only for systems with an explicit query/lock (materialization) interface; for opaque learned policies the stages are not individually recoverable, so the framework is *complementary to*, not a replacement for, probing, causal-mediation, and message-ablation diagnostics. On two local LLM agents (qwen3:4b, llama3.1 on TextNav), the same logger localizes a completion-limited failure: under a tight step budget both agents form the query, retrieve, and lock the target ($Q^{\star}=R^{\star}=M^{\star}=1$) yet none complete ($C^{\star}=0$), so an otherwise opaque

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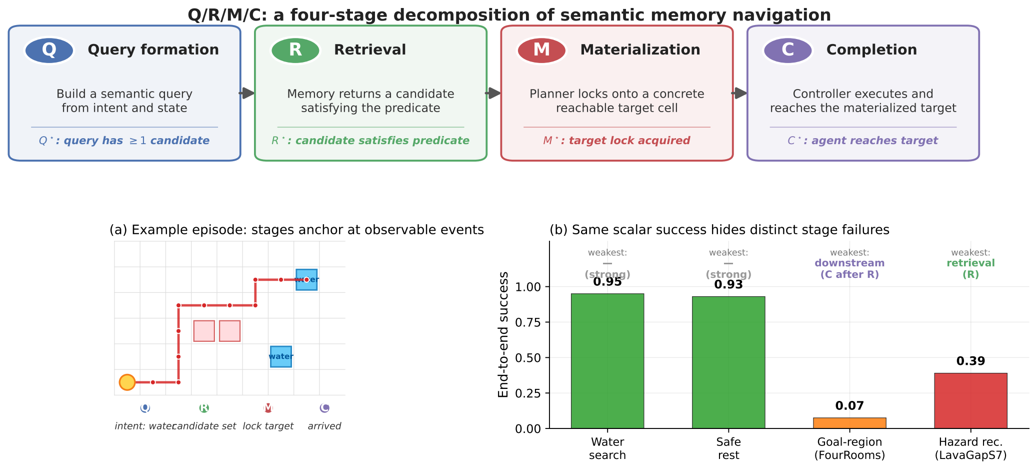

Figure 1: Q/R/M/C decomposes one semantic-navigation episode into four observable stages. **Top:** the four stages and their event-level definitions. **(a)** A successful water-search episode anchors at one event per stage, producing logged starred events $Q^\star, R^\star, M^\star, C^\star$. **(b)** The same scalar "end-to-end success" summary masks qualitatively different failures on the four single-agent task families: water and rest are strong; goal-region rejoin (FourRooms) is downstream-limited; hazard recovery (LavaGapS7) is retrieval-limited (oracle interventions, §3.4).

success collapse is resolved into a specific failing stage rather than a leaderboard number. Thus the contribution is a measurement protocol for making memory failures comparable across single- and multi-agent symbolic and LLM systems wherever a query/lock interface is exposed, and for measuring stage observability itself on opaque learned policies. **Scope.** The primary experiments are in grid-world substrates (MiniGrid, BabyAI, a custom 12×10 grid). The contribution is the diagnostic framework and its validation; a continuous-state validation on VMAS (three scarcity cells $\times$ 40 seeds) reproduces both slope signs with block-bootstrap CIs excluding zero (Spearman -0.999 and $+0.995$; Appendix F.7); visually rich 3D substrates (Habitat, ProcTHOR) remain future work.

1 Introduction

Classical navigation benchmarks often assume that the agent is given an explicit destination coordinate or a dense task reward. Many practical navigation problems are different: the agent must reach a place of a certain *type*, such as a safe rest area, a water-like resource, a post-hazard recovery exit, or a goal-associated region, specified by a semantic pattern rather than a fixed (x, y) waypoint. The place must therefore be *remembered and retrieved by meaning*. This setting opens a methodological question: can we measure *where* memory-based semantic navigation breaks down, not only *whether* it succeeds?

We answer this question with a four-stage diagnostic framework that applies to both single agents and multi-agent collectives. The decomposition is Q/R/M/C: **Q**uery formation, **R**etrieval satisfaction, target **M**aterialization, and **C**ompletion after retrieval. The decomposition factorizes per-episode semantic-path completion and, under a conditional stage-Markov property, its factors are estimable from trajectory logs (Definition 1). The rest of the paper develops a single argument in three parts.

Single agent. Stage attribution is non-trivial. Success-only metrics collapse qualitatively different failures into a single scalar. In a 10-seed MiniGrid benchmark, water and rest tasks are strong end-to-end (0.95, 0.93 success), but goal-region rejoin and hazard recovery are substantially harder, and for *different reasons*. Oracle-stage interventions show that hazard recovery is retrieval-limited (oracle retrieval lifts success $0.39 \to 0.60$), while goal-region rejoin is downstream-limited (oracle retrieval rescues semantic-path completion without changing end-to-end success). Inspection of the framework's own logs then sharpens the attribution: the retrieval bottleneck is not candidate *ranking* (precision 96.9% when candidates exist) but candidate *generation*: the memory contains no eligible symbols at 91% of decision points. The single-agent bottleneck is a *memory-accumulation* limit: evidence is observed but never consolidated into queryable structure.

Multi-agent. Consolidation acquires a rate. When N agents share memory through a broadcast protocol with cadence K, the factorization extends with enlarged conditioning sets (Definition 2), and K becomes a structural axis: it is, mechanically, a knob on *how fast private memories consolidate into a collective one*. A slope-level claim (Empirical Claim 1) predicts a bottleneck shift: as K decreases, $P(M^\star|R^\star)$ rises (fresher peer memory enables reliable commitment) and $P(C^\star|M^\star)$ falls (peers race for the same targets). Both slope signs are confirmed on a 35,640-run sweep as moderate, cluster-robust effect sizes (Appendix F.3), the shift is directly visible as a negative within-cadence correlation between the two links, and the resulting trade-off produces an interior optimum of mean time-to-success at $K=8$: consolidation has a non-trivial optimal rate.

Diagnostic synthesis. Both regimes point at the same measurement object. The single-agent diagnosis says that evidence consolidation into symbols can be the binding constraint; the multi-agent diagnosis says that the rate and cost of sharing consolidated memory determine the optimum. Q/R/M/C is the common instrument: candidate systems can be compared by how they move $P(R^\star|Q^\star)$, $P(M^\star|R^\star)$, and $P(C^\star|M^\star)$, even when their end-to-end success rates look similar.

> **Scope and non-claims (consolidated).** All caveats of this paper in one place; the running text no longer repeats them. Each limitation is stated together with the reason the results survive it. (i) The contribution is the operationalisation: log-level definitions under which the four conditional rates are identifiable on any system exposing a query/lock interface (Definitions 1–2). The consistency argument itself is the chain rule plus frequency estimation, and no mathematical novelty is claimed for it. (ii) Empirical Claim 1 is a slope-level empirical observation rather than a theorem, stated deliberately so that each slope is separately falsifiable on logs; both slopes were confirmed with cluster-robust confidence intervals (Appendix F.3). (iii) We make no claim about the shape of the conditional product Φ; in our data it is monotone in K, and the interior optimum lives on mean time-to-success (§4.5). (iv) Primary substrates are grid worlds because they admit oracle interventions and exhaustive sweeps that richer substrates do not; VMAS is a directional continuous-state check, and the LLM studies are small and controlled (two local models on TextNav; an N=3 collective; a three-framework external validation at n=20 per cell with two models, §5). (v) Multi-agent place identity assumes a shared coordinate frame for the main sweep (relaxed to semantic, frame-free matching in Appendix E.3); the assumption affects the agents' merge, not the event definitions. (vi) The four-rate product converges to *semantic-path completion* $C^\star$, not to raw task success Y: Y can exceed $C^\star$ when an episode succeeds without traversing the retrieval path. This is by design — Q/R/M/C localises where the *semantic route* fails, which a success-only metric conflates with every other route to success. (vii) Stage-separability requires an explicit query/lock interface; on opaque learned policies the stages are not individually observable and the framework is *complementary to*, not a replacement for, probing, causal-mediation, and message-ablation diagnostics. A stage-level operationalisation is testable on specific summaries and can be *wrong* on others — (iii) is an instance we report — which is what separates a measurement protocol from a stage-agnostic guess.

Why this framing is timely. Embodied and goal-conditioned agents increasingly retrieve places, objects, or tools by meaning (Andrychowicz et al., 2017; Ghosh et al., 2021; Savva et al., 2019). LLM-agent systems lean on explicit memory layers but evaluate mostly end-to-end (Wang et al., 2023; Shinn et al., 2023; Packer et al., 2023). Multi-agent reinforcement learning likewise increasingly involves communicating populations (Lowe et al., 2017; Foerster et al., 2018), but the question "which agent should know what, and when?" lacks a unified diagnostic vocabulary. Our framework supplies one.

Contributions. This is a methodology paper. We are explicit about what is bookkeeping vs. empirical claim vs. instantiated artefact.

1. **Measurement protocol (single-agent).** The Q/R/M/C decomposition is operationalised so that the conditional rates are estimable from trajectory logs (Definition 1); the contribution is the *logging discipline* that makes the decomposition usable on standard navigation traces (non-claims: see box above).

Table 1: One contract, five system classes. The same four events ($Q^\star, R^\star, M^\star, C^\star$), defined once, instrument each class without modification — and localize a different failure in each.

System class	Substrate	What the same four events revealed
Symbolic single agent	MiniGrid, BabyAI	ranking near-optimal (96.9%) while candidate generation is empty at 91% of decisions: a memory-*accumulation* limit (§3)
Symbolic collective (4 architectures $\times$ cadence K)	grid, 35,640 runs	M↔C bottleneck shift; interior t_{succ} optimum at $K{=}8$; provenance mechanism (§4)
Learned policies (BC; CommNet-PPO)	grid	competitive success with structurally collapsed stage observability (§3, §4)
LLM agents (2 models; $N{=}3$ collective)	TextNav	an opaque $1.00 \rightarrow 0.00$ collapse resolved to pure C-budget failure; sharing lowers success $2/3 \rightarrow 1/3$ (§4)
Third-party stacks (OpenAI SDK, LangGraph, AutoGen) $\times$ 2 models	tool traces	structural failures localize identically (20/20); first-commitment accuracy separates models; frameworks reshape it (§5)

2. **Measurement protocol (multi-agent).** An extension to N communicating agents with two coupling channels (memory $\mathcal{M}_{-i}^{(K)}$, occupancy $\mathcal{O}_{-i}$) under a peer-broadcast protocol with cadence K (Definition 2).
3. **Empirical Claim 1 (bottleneck shift).** On a 35,640-run multi-agent sweep, the two stylised slope conditions **(F)** and **(O)** are both confirmed as moderate, cluster-robust effect sizes (Spearman -0.53 and $+0.43$ vs. K, 95% CIs excluding zero over 81 configuration cells; Appendix F.3; full benchmark in §4.5), with a provenance-level mechanism (§4.5).
4. **System and benchmarks.** A symbolic-semantic memory stack with four communication architectures (independent, shared-bus, central-aggregator, peer-to-peer broadcast); single-agent oracle benchmark (10-seed MiniGrid, BabyAI portability); 35,640-run multi-agent sweep at $N \in \{3, 5, 8\}$ + 8,640-run scale-up at $N = 12$; 100-run portability sweep on a standard-MiniGrid wrapper.
5. **Diagnostic comparison across architectures.** The same logging contract separates symbolic memory systems from learned policies with implicit communication: the latter can be competitive on success while collapsing Q/R/M/C stage observability. This is an evaluation result, not a claim that one architecture class should always dominate.
6. **Scope and honesty.** The primary experiments are in grid-world substrates. We treat that as a methodology-paper choice and state it explicitly (§6); a continuous-substrate validation on VMAS with cluster-robust CIs is included (Appendix F.7); visually rich 3D substrates (Habitat, ProcTHOR, MineDojo) are future work.

Portability overview. The central claim is not any single number below; it is that one logging contract transfers across system classes and returns a located diagnosis in each. Table 1 summarises this correspondence; each subsequent section instantiates one row.

2 The Q/R/M/C decomposition

2.1 Problem formulation and notation

The agent operates in a partially observable navigation environment. At each step it receives an observation o_t, maintains a memory state $\mathcal{M}_t$, and forms a query q_t encoding the current task intent. A retrieval step returns a candidate set of places matching q_t from $\mathcal{M}_t$; a chosen candidate is materialized into an actionable target; a local controller executes actions until either the target is reached or the episode times out. The target is not a coordinate but a semantic pattern, so success is a function of the chosen candidate's semantic profile, not of any pre-supplied coordinate.

Habitat, AI2-THOR, and ProcTHOR made visually rich navigation evaluation standard (Savva et al., 2019; Kolve et al., 2017; Deitke et al., 2022). We deliberately focus on MiniGrid-class environments to obtain clean and reproducible stage measurements, and use BabyAI only as a portability check. We do not claim to solve LLM-agent memory here; rather, we argue that the same Q/R/M/C decomposition is a useful template for evaluating memory-grounded agent decisions beyond navigation.

The central empirical question is therefore not just whether the agent succeeds, but *which stage fails first* when it does not. Two related measurement layers are used. **Operational rates** count, per query attempt, whether the query yielded a non-empty candidate set, whether the chosen candidate satisfied the semantic predicate, and whether it was materialized into an actionable target; *completion after retrieval* is an episode-level indicator. **Nested episode-level events** $Q^\star \Rightarrow R^\star \Rightarrow M^\star \Rightarrow C^\star$ are used in the factorization below and audited in §3.2. Full operational definitions are given in Appendix C.

Notation. The four stages are Q, R, M, C with episode-level starred events $Q^\star, R^\star, M^\star, C^\star \in \{0,1\}$, nested by construction. $P(\cdot)$ denotes the population probability; $\hat{P}(\cdot)$ its empirical-frequency estimator. **The letter M always refers to the Materialization stage; the number of targets in the multi-agent environment is written T (not M, constraint $T \le N$, "scarcity" = $N > T$).** Other symbols: $\mathcal{C}_K$ peer-broadcast protocol with cadence K, $\mathcal{M}_{-i}^{(K)}$ peer memory snapshots, $\mathcal{O}_{-i}$ peer occupancy, $\nu(\cdot)$ freshness functional, $\Phi(K) = P(M^\star | R^\star; K) \cdot P(C^\star | M^\star; K)$, t_{succ} mean time-to-success. Full symbol table in Appendix D.1.

2.2 Single-agent factorization

Let Y denote overall episode success, and let $Q^\star, R^\star, M^\star, C^\star$ denote the nested episode-level semantic-path events. The framework factorizes *semantic-path completion* rather than raw task success:

$$Q^\star \to R^\star \to M^\star \to C^\star.$$

For any query q and memory state $\mathcal{M}$,

$$\begin{aligned} P(C^\star = 1 \mid q, \mathcal{M}) = P(Q^\star = 1 \mid q, \mathcal{M})\; P(R^\star = 1 \mid Q^\star = 1, q, \mathcal{M}) \\ \cdot\, P(M^\star = 1 \mid R^\star = 1, q, \mathcal{M})\; P(C^\star = 1 \mid M^\star = 1, q, \mathcal{M}). \end{aligned}$$

Overall task success Y can exceed $C^\star$ because some episodes succeed without traversing the full semantic retrieval path.

Assumption 1 (conditional stage-Markov property). Conditioned on the current stage being successful, the probability of the next stage depends only on the current successful stage, the current query, and the current memory state. Formally,

$$P(R^\star \mid Q^\star, q, \mathcal{M}, \text{history}) = P(R^\star \mid Q^\star, q, \mathcal{M}),$$

and analogously for $M^\star$ and $C^\star$.

Assumption 2 (positivity). Each conditioning event used above has non-zero probability on the support of the benchmark.

Definition 1 (single-agent stage estimability). Under Assumptions 1–2, the four conditional rates in the factorization above are well-defined population quantities and the empirical-frequency estimators are consistent estimators of them; their product converges to semantic-path completion. The argument is elementary — the chain-rule factorization plus consistency of frequency estimators on Bernoulli events — and the value of stating it is operational, not mathematical: it pins down the *logging discipline* (which events, conditioned on what) under which the decomposition is readable from standard trajectory logs, and it is the contract that every experiment in this paper audits against (§3.2). Full statement and elementary argument in Appendix A.

When Assumption 1 may fail. Assumption 1 holds tightly here because q and $\mathcal{M}$ are explicitly logged at every stage event. Two practically relevant violations remain: adaptive in-episode query rewrites not surfaced to the logger, and hidden between-stage memory updates. In those cases the factorization should be read as a projection onto observable traces rather than as a full structural identification claim. Appendix F.6 quantifies the estimator bias under a controlled violation of this kind: it is negligible for mild unlogged couplings and one-signed (an upward bias on $P(C^\star | M^\star)$), which bounds the framework's exposure when it is applied to learned or LLM agents.

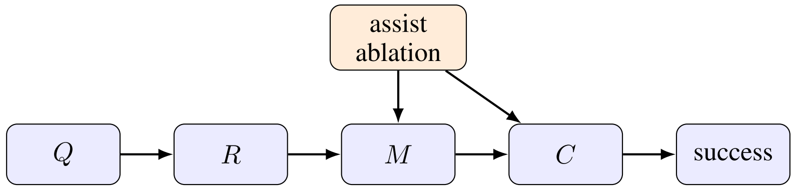

Figure 2: Stage-structured interpretation of the Q/R/M/C framework. Assist on/off ablations target intermediate mechanisms; in the multi-agent extension the cadence protocol $\mathcal{C}_K$ acts on the R→M and M→C edges.

Mechanism-level reading. The same decomposition admits a stage-structured mechanism interpretation. Figure 2 treats stage outcomes as a directed acyclic graph. Under this interpretation, assist on/off ablations are mechanism-targeted perturbations of materialization and post-retrieval execution rather than undifferentiated sensitivity analyses; in the multi-agent setting the same graph is replicated per agent and connected through the cadence protocol $\mathcal{C}_K$, which acts on $\mathcal{M}_{-i}^{(K)}$ at the R→M edge and on $\mathcal{O}_{-i}$ at the M→C edge — exactly the two edges where Empirical Claim 1 will predict the slope shift.

Relation to prior work. Closest are (i) cognitive-map and semantic-map frameworks (Tolman, 1948; O'Keefe and Nadel, 1978; Kuipers, 2000; Gupta et al., 2017; Savinov et al., 2018), (ii) goal-conditioned and successor-representation RL (Andrychowicz et al., 2017; Ghosh et al., 2021; Dayan, 1993; Stachenfeld et al., 2017; Momennejad et al., 2017), (iii) multi-agent communication (Lowe et al., 2017; Foerster et al., 2018; Sukhbaatar et al., 2016; Foerster et al., 2016) and gossip/consensus (Boyd et al., 2006; Xiao et al., 2007), and (iv) LLM-agent memory layers evaluated end-to-end (Wang et al., 2023; Shinn et al., 2023; Packer et al., 2023). Our framework is orthogonal to all four: it *diagnoses* where in the Q/R/M/C chain a given architecture succeeds or fails and lifts the cadence axis from network protocols to a cognitive-architecture parameter (full discussion in Appendix F.11).

2.3 Event semantics, bounded rationality, and an order-theoretic reading

The four starred events anchor at concrete logged triggers. We state the triggers once, because every empirical claim in the paper is a claim about them (per-regime formal definitions in Appendix C), and we give each stage its reading in terms of the resource it bounds.

- $Q^\star$ **(intent issued).** The planner emits a semantic intent $q = (q^{\text{req}}, q^{\text{pref}}, q^{\text{pen}})$: small sets of required, preferred, and penalty tags. The episode-level event fires on emission, so it is saturated whenever the task is semantic ($\hat{P}(Q^\star)$=1.00 in Table 2); the multi-agent per-tick logger additionally requires a non-empty candidate set (Appendix C.1), a bookkeeping difference we record rather than hide. Resource bound: attention — the intent is a few tags, not the full state.
- $R^\star$ **(memory answers).** Retrieval evaluates q against the memory and returns a candidate set; the event fires when the set contains a candidate satisfying the task predicate (within ϵ of a true target in the multi-agent logger). The 91% empty-candidate-set failures of §3.5 are R-side events: no stored place clears the required-tag threshold. Resource bound: memory — retrieval can only surface what consolidation has stored.
- $M^\star$ **(target lock).** The planner materializes one candidate into a concrete actionable cell. Candidacy itself is threshold-gated (a place enters the candidate set only if its score clears the planner's gate, Appendix D), and the lock commits to a single candidate once per query rather than re-optimizing against the full memory at every tick. The event fires on lock acquisition (within ϵ of a true target in the multi-agent logger). Resource bound: deliberation.
- $C^\star$ **(closed-loop completion).** The local controller executes until the locked cell is reached or the episode times out; the event fires on arrival. C is the only stage whose failure modes are extra-semantic: controller competence and, in the multi-agent regime, occupancy $\mathcal{O}_{-i}$. Resource bound: control, and in a collective, shared targets.

Observable-interaction reading. The four events admit a uniform reading as semantically distinct acts in the externally observable interaction, with no reference to internal computational state: $Q^\star$ is an informational commitment (a query stating what evidence is needed); $R^{\star}$ is the memory's answer to that commitment; $M^{\star}$ is a public commitment to act on specific evidence, existing exactly from the moment the lock is written — and, in a collective, visible to peers and occupying a target; $C^{\star}$ is the environment's confirmation that the commitment was realized. This grounds Materialization independently of any cognitive interpretation (the appearance of a public commitment is an objective log event), and such commitments are the source of the occupancy contention of §4.5. The stance mirrors runtime verification and distributed-system observability, where specifications constrain externally visible events rather than internal state (Leucker and Schallhart, 2009).

Interpretive readings (all deferred). The stages admit three readings, none of which is a contribution and all of which now live in the appendices: the pipeline as a *satisficing, boundedly rational* agent whose four resource bounds (attention, memory, deliberation, control) are exactly the four failure loci (Appendix B); an *order-theoretic* account of empty retrieval — the 91% regime — via the query–extent Galois connection (Appendix A.4); and an *options/semi-MDP* account of why optima appear on time-bearing metrics rather than on the duration-blind chain probability Φ (Appendix B).

3 Single agent: stage diagnostics localize failure

This section instantiates the first row of Table 1: a single symbolic agent, the simplest system class, where the question is whether stage attribution adds information that end-to-end success does not.

3.1 System, tasks, and protocol

Memory stack. The system uses a graph substrate with a semantic retrieval interface on top: observations o_t are tokenized into symbolic scene tokens and semantic tags that update a memory of place records, transitions, episodic traces, and concept/relation records. Planner queries return candidates by a weighted log-score over required, preferred, and penalty tags. The retrieval machinery is reused across all tasks and architectures; full memory schema, update rules, and scoring formula are in Appendix D.

Tasks, baselines, protocol. Four single-agent task families are evaluated (water, safe-rest, goal-region on Empty-6x6 / FourRooms (Sutton et al., 1999), hazard recovery on LavaGapS7), 10 seeds $\times$ 8 episodes each, $\times$ 2 assist modes. Semantic stack (node, concept, state-conditioned variants) is compared against four baselines: random, graph-only, SPTM-like, learned BC. Reported: end-to-end success, steps, four operational stage rates, assist deltas, weakest stage per task and per method. All means are seed-aggregated; 95% bootstrap CIs use $B{=}5{,}000$ resamples (Efron, 1979); assist comparisons use paired Wilcoxon signed-rank tests (Wilcoxon, 1945) with Holm-Bonferroni correction (Holm, 1979) (Appendix G). For the two hardest tasks (FourRooms, LavaGapS7), four oracle regimes (*base*, *oracle R/M/C*) are run — each replaces exactly one stage while leaving the rest of the pipeline unchanged. BabyAI portability check in Appendix E. All single-agent experiments ran on a 10-core ARM CPU, 32 GB RAM (CPU-only); oracle sweep < 1 min wall-clock.

LLM diagnostic block. To check that Q/R/M/C is not tied to the symbolic planner implementation, we also run a fixed TextNav substrate with an LLM-driven observe/query/retrieve/decide loop. The TextNav task, prompts, parser contract, and event definitions are held fixed while only the local Ollama model changes; the backend uses model-compatible completion settings when an Ollama chat template otherwise hides short structured answers in a separate thinking field. This block is a controlled step-budget study (Table 4): sweeping the completion budget drives both LLM agents into a failure that Q/R/M/C localizes to the C stage while $Q/R/M$ stay saturated, exhibiting the framework's diagnostic property on real language agents. Reproducing full ReAct/Reflexion/Voyager/MemGPT-style agents is a separate systems contribution; the comparison to those systems is a coverage comparison in Appendix F.12.

3.2 Stage-estimator validation

Before the main benchmark, the factorization is validated. On a synthetic four-stage process with known probabilities the nested-estimator product tracks empirical semantic-path completion across

Table 2: Stage-factor consistency audit for the best semantic method in each task family. The product of nested conditional estimators matches semantic-path completion; overall success can be higher because some episodes succeed without traversing the full semantic retrieval path.

Task	Method	Overall	SP-comp	$\hat{P}(Q^\star)$	$\hat{P}(R^\star \mid Q^\star)$	$\hat{P}(M^\star \mid R^\star)$	$\hat{P}(C^\star \mid M^\star)$	Product
Water	`water-cp`	0.95	0.70	1.00	0.98	0.74	0.97	0.70
Rest	`rest-s2`	0.93	0.73	1.00	1.00	0.80	0.91	0.73
Goal region	`goal-n1`	0.54	0.00	1.00	0.54	0.01	0.00	0.00
Hazard recovery	`hazard-s7`	0.40	0.16	1.00	0.56	0.58	0.50	0.16

Table 3: Baseline comparison. The learned behavior-cloned baseline dominates the easiest resource tasks and hazard recovery, while the semantic stack remains strongest on the relation-heavy goal-region family and remains the only family with interpretable Q/R/M/C structure.

Task	Random	Graph-only	SPTM-like	Learned	Best semantic
Water	0.85	0.90	0.90	1.00	0.95
Rest	0.85	0.90	0.90	1.00	0.93
Goal region	0.41	0.49	0.52	0.50	0.54
Hazard recovery	0.08	0.39	0.00	0.74	0.40

sample sizes. On the real benchmark, the same product matches semantic-path completion rather than overall success (Table 2) — exactly the contract of Definition 1.

3.3 Success-only summary and what it hides

Water and rest are strong end-to-end tasks (0.95 and 0.93 success; Figure 1b). Goal-region reaches 0.54 success at the task-family level — an average that pools a solved environment (Empty-6x6, 1.00) with a hard one (FourRooms, 0.075; §3.4), so the family number is bimodal rather than a uniform mid-difficulty — but the bottleneck sits downstream of candidate selection (§3.4). Hazard recovery is the hardest retrieval-sensitive task at 0.40 success, with retrieval satisfaction as the weakest stage. Under success-only evaluation the two hard families look like the same kind of weakness; the stage view will show they are opposites. Per-task operational rates with 95% CIs are in Table 8 (Appendix D.3).

Baseline comparison. Table 3 compares the semantic stack against random, graph-only, SPTM-like, and learned behavior-cloned baselines. The learned baseline dominates the easiest resource tasks (1.00 on water and rest) and hazard recovery (0.74), while the semantic stack remains strongest on the relation-heavy goal-region family (0.54). The relevant contrast for this paper, however, is not the success column: the semantic stack is the only method family whose decisions expose interpretable Q/R/M/C structure. On the BC baseline the stage events are degenerate for the same structural reason as on the multi-agent learned baseline analysed in §4.5 — a policy without an explicit query/lock interface cannot emit separable stage events, so the diagnostic decomposition has nothing to attach to. End-to-end learning and stage diagnostics answer different questions; this paper is about the second.

3.4 Oracle-stage interventions localize the bottleneck

On goal-region the semantic variants solve Empty-6x6 (1.00 success) but collapse on FourRooms (0.075). For hazard recovery (LavaGapS7), base success 0.39 (bootstrap 95% CI $[0.35, 0.43]$) rises only to 0.43 under oracle controller, but jumps to 0.59 under oracle materialization and 0.60 under oracle retrieval (CI $[0.51, 0.69]$; per-seed paired deltas positive in $9/10$ seeds, exact sign test $p{=}0.004$); goal-region rejoin is downstream-limited (oracle R lifts semantic-path completion to 0.08 without moving end-to-end success). The per-task attribution as a stacked waterfall is in Figure 6 (Appendix D.3). Two tasks identical under success-only evaluation fail at opposite ends of the Q/R/M/C chain.

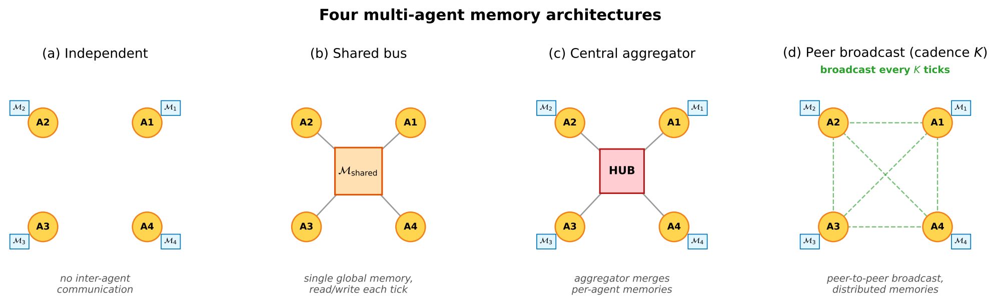

Figure 3: Four multi-agent memory architectures. **(a)** Independent: per-agent private memory, no exchange. **(b)** Shared bus: one global memory, every agent reads/writes each tick. **(c)** Central aggregator: per-agent memories plus a hub that merges them into a consensus report. **(d)** Peer-to-peer broadcast: per-agent memories with pairwise snapshot exchange every K ticks (the cadence axis of Empirical Claim 1).

3.5 The logs refine the diagnosis: a memory-accumulation limit

Inspecting the framework's own logs sharpens the retrieval attribution. On the 10-seed hazard-recovery run, 349 retrieval calls were logged: only 32 (9.2%) returned at least one candidate, and of these 31 (96.9% precision) selected a semantically satisfied candidate. The bottleneck is therefore not candidate *ranking* but candidate *generation* — the symbolic memory does not contain hazard-recovery-tagged nodes at 91% of decision points. A retrained 2-layer candidate-generation MLP trained on per-state features (test accuracy 0.59 vs majority-class 0.81; Appendix H) confirms that instantaneous state features cannot substitute for the missing structure. The framework's "retrieval-limited" attribution thus refines to a *memory-accumulation* limit: candidate generation cannot be solved from state features alone, and the binding constraint is how trajectory evidence is consolidated into persistent symbolic structure. This attribution is not an artefact of the required-tag threshold θ: sweeping θ across the full range leaves the empty-candidate-set rate invariant (Appendix F.5), because the empties are absence of any stored candidate, not sub-threshold filtering. Nor is it an artefact of the weighting: the empty rate is 0.908 pooled over calls, 0.909 seed-weighted, and 0.891 episode-weighted (mean over the 80 episodes of each episode's own empty share), so long failing episodes contributing more decision points do not inflate the figure. Accumulation itself is not one of the four stages: the protocol localised the failure to the R link, and this within-link forensics named the cause — we read Q/R/M/C as coarse localisation with a designated drill-down, not as a complete causal ontology of memory failure. This sets up the multi-agent section: when consolidation becomes a shared process across N agents, it acquires a measurable *rate*.

4 Multi-agent: memory as a shared resource

This section instantiates rows two to four of Table 1: the same logged events, unchanged, instrument a communicating collective — and, within the same section, learned policies and LLM agents — where the additional failure axis is contention for the targets that memory designates.

4.1 Four memory architectures and the cadence axis

Four communication architectures are instantiated; they share the single-agent retrieval substrate but differ in how memory is distributed across agents (Figure 3).

Independent. Each agent maintains a private event store and concept graph. No inter-agent communication mechanism exists. Lower-bound baseline.

Shared bus. All agents publish observations to a single *collective memory* event bus. One shared concept graph is rebuilt from the union of events. Maximally centralized.

Central aggregator (ConsensusLayer). Each agent has its own private concept graph; a central layer pulls snapshots and computes a trust-weighted merge, producing a single *consensus report* consumed by all agents.

Peer-to-peer broadcast (cadence K). Each agent has its own concept graph and its own merged view. Every K ticks each agent broadcasts a snapshot to a passive bus; each agent independently merges its own snapshot with the latest snapshot received from each peer. Different agents may hold different merged views; there is no central report.

The first three correspond to the three regimes in the multi-agent communication literature: no exchange, CTDE-style aggregation, and decentralized peer communication. The fourth is the architecture for which Definition 2 and Empirical Claim 1 are directly applicable, since the cadence K is a tunable parameter rather than a fixed protocol. Mechanically, K is the rate at which private memories consolidate into a collective view, which is why it is the structural axis of this section.

4.2 Multi-agent factorization

Extend the factorization to a population of N agents that share information through a communication protocol $\mathcal{C}_K$ parameterized by broadcast cadence K. Each agent i has its own memory state $\mathcal{M}_i$ and issues its own query q_i. Let $\mathcal{M}_i^{(K)}$ denote the projection of agent i's memory at the moment it queries, after absorbing every peer message received under cadence K. Two coupling channels are separated: (i) memory coupling $\mathcal{M}_{-i}^{(K)}$, the latest peer memory snapshots accessible to agent i; (ii) occupancy coupling $\mathcal{O}_{-i}$, the set of targets that other agents have already claimed by the time i acts.

Definition 2 (multi-agent stage estimability). For each agent i,

$$P(C_i^\star \mid q_i, \mathcal{M}_i, \mathcal{C}_K) = P(Q_i^\star \mid q_i, \mathcal{M}_i)\; P(R_i^\star \mid Q_i^\star, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)})$$
$$\cdot\, P(M_i^\star \mid R_i^\star, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)})\; P(C_i^\star \mid M_i^\star, q_i, \mathcal{O}_{-i}).$$

Under a multi-agent stage-Markov assumption (Appendix C), the four factors are estimable from per-agent trajectory logs and the empirical-frequency estimators are consistent.

When Definition 2 may fail. The single-agent stage-Markov property absorbs trajectory history into $(q, \mathcal{M})$. In the multi-agent setting, the analogous absorption requires that dependence on past peer decisions enters only through $\mathcal{M}_{-i}^{(K)}$ at the M stage and through $\mathcal{O}_{-i}$ at the C stage. This step is non-trivial: $\mathcal{O}_{-i}$ summarises target occupancy at decision time but loses information about *how* peers arrived there. The factorization is therefore a projection onto observable per-agent traces under that summary. Two specific failure modes arise: (i) inter-agent communication side-channels not logged through $\mathcal{C}_K$; (ii) coordinated peer policies whose decision at the C stage depends on hidden peer state not summarised by $\mathcal{O}_{-i}$.

4.3 Bottleneck shift

Empirical Claim 1 (bottleneck shift M↔C, slope-level). Assume scarcity ($N > T$ targets, see §2.1) and a peer-broadcast architecture with cadence $K \in \mathbb{N}$. Two stylized assumptions are posited: **(F)** memory freshness $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise as K decreases, and the probability that agent i's planner locks onto a reachable target is non-decreasing in ν; **(O)** the probability that an arbitrary committed target is already in $\mathcal{O}_{-i}$ is non-increasing as K increases. Write $P(M^\star|R^\star; K)$ and $P(C^\star|M^\star; K)$ for the population-level conditional probabilities (averaged over agents and the seed distribution of the benchmark); $\hat{P}(\cdot)$ denotes the corresponding empirical-frequency estimator computed from logs. Under (F) and (O),

$$\frac{\partial\, P(M^\star|R^\star; K)}{\partial K^{-1}} > 0, \qquad \frac{\partial\, P(C^\star|M^\star; K)}{\partial K^{-1}} < 0.$$

Both slope conditions are direct empirical predictions that are testable on logs through $\hat{P}$.

What this claim is and is not. Empirical Claim 1 is a *slope-level statement on conditional rates*: each of the two slopes is testable and is confirmed on the 35,640-run sweep (§4.5) and independently on a 100-run standard-MiniGrid portability sweep (§6). We report the slopes as *effect sizes with cluster-robust confidence intervals* (resampled over the 81 independent configuration cells rather than the correlated run count) in Appendix F.3. The claim's boundary with respect to the conditional product Φ is stated once, in the Scope-and-non-claims box (§1) and in §4.5. The shift is a signature of scarce-target contention, not of the grid design: under abundance ($T>N$) both slopes flatten and the M$\leftrightarrow$C anticorrelation attenuates (Appendix F.9), i.e. the prediction breaks exactly where the contention mechanism is removed.

Chronology. Empirical Claim 1 is post-hoc. It was formulated *after* an initial 12,960-run sweep conducted under four stage-agnostic hypotheses (Appendix F); the two slope conditions were then confirmed on the full 35,640-run sweep, on the $N=12$ scale-up (Appendix E), and on the MiniGrid portability wrapper (§6). We state this explicitly: the verified slope signs are predictions on enlarged data, not a pre-registered (externally timestamped) prediction. A fully formal version is left for future work.

4.4 Sweep protocol

The four architectures are evaluated on a custom multi-agent grid (12×10 cells); each agent must reach a cell tagged `water_source`, choosing actions via a shared memory-driven planner so only the memory layer differs. The sweep crosses $N\in\{3,5,8\}$, $T\leq N$, three layouts (symmetric/asymmetric/random), three hazard densities, four architectures, eight peer cadences $K\in\{1,2,4,8,16,32,48,64\}$, 40 seeds — 35,640 runs in ≈ 22 min on 8 cores. Per-agent per-tick Q/R/M/C indicators are logged (Appendix C); episode-level starred events are the disjunction over ticks. A separate oracle-intervention block on scarcity cases $(N,T)\in\{(3,2),(5,3),(8,5)\}$ at five cadences evaluates oracle R (memory returns ground-truth water positions), oracle M (planner locks onto nearest real water), and oracle C (completion of a real materialised lock is guaranteed; Appendix F.14).

4.5 Results: stage decomposition and bottleneck shift

The 35,640-run sweep is reported through four observations, structured to mirror Empirical Claim 1: (i) the conditional decomposition isolates the M and C stages as the differentiating links; (ii) the bottleneck shift appears directly as a negative within-cadence correlation between $P(M^\star|R^\star)$ and $P(C^\star|M^\star)$, peaking at $K=4$ and decaying to noise at $K=64$; (iii) on mean time-to-success, the resulting trade-off produces an interior optimum at $K=8$ — robust in location but shallow in margin, with the advantage over the adjacent fast cadences not resolved at this sample size (Appendix F.3); (iv) on the conditional product Φ, the same trade-off produces a monotone trend rather than an interior peak — a summary statistic on which we make no formal claim. A learned-communication baseline and a multi-agent oracle-intervention loop close the subsection.

4.5.1 Conditional decomposition reveals where each architecture fails.

Figure 4 shows per-architecture conditional Q/R/M/C rates; full numerics with bootstrap CIs are in Appendix F.1, Table 11. Retrieval conditional $P(R^\star|Q^\star)$ is saturated; the differentiating signal lives at M and C. Shared-bus, central-aggregator, and fast-broadcast peer ($K\leq 4$) have high $P(M^\star|R^\star)$ but low $P(C^\star|M^\star)$; independent and slow-broadcast peer ($K\geq 16$) show the reverse — the M$\leftrightarrow$C trade-off predicted by Empirical Claim 1.

4.5.2 Bottleneck shift and interior efficiency optimum.

Across all peer runs, Spearman of $P(M^\star|R^\star)$ vs K is -0.53 (95% cluster-robust CI $[-0.59,-0.47]$) and of $P(C^\star|M^\star)$ vs K is $+0.43$ (CI $[+0.37,+0.50]$); both are moderate effect sizes whose CIs exclude zero under resampling of the 81 configuration cells (Appendix F.3), matching the slope conditions of Empirical Claim 1. We report these rather than p-values, which are uninformative at this correlated sample size. Both slope signs are also robust to the two free thresholds of the logger — candidacy θ and lock tolerance ϵ (Appendix F.5). The stronger empirical signature is at the per-configuration level: Pearson of $P(M^\star|R^\star)$ vs $P(C^\star|M^\star)$ *within* a fixed K is $-0.43,-0.54,-0.61,-0.19,-0.05,-0.04,-0.02,-0.02$ at K=$1,2,4,8,16,32,48,64$, peaking at K=4 and decaying to noise by K=64 (Figure 9, Appendix F.1). Configurations where M is high systematically have low C: the bottleneck migrates between the two stages within the same $(N,T,\text{layout},\text{hazard},\text{seed})$ cell as K varies. On the efficiency metric, mean t_{succ} has a U-shape with interior minimum at K=8 (7.47 ticks vs 8.02 for the independent baseline); the minimum's

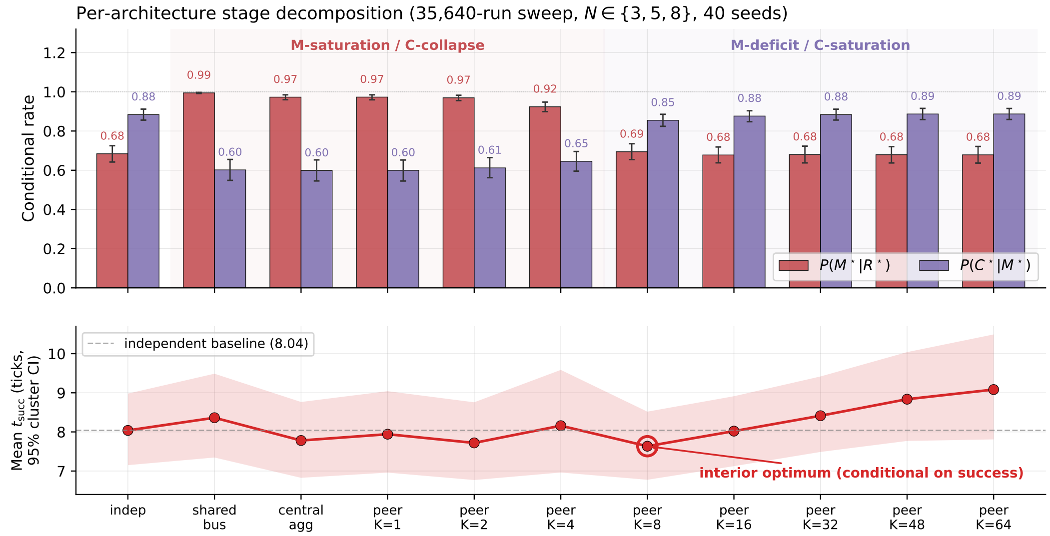

Figure 4: Per-architecture conditional rates (top) and mean time-to-success among successes (bottom) from the 35,640-run sweep. Error bars and band: 95% cluster-bootstrap CIs over the 81 design cells; points are cell-weighted means (7.63 at K=8 vs. 8.04 independent; the run-pooled values of Table 11 are 7.47/8.02). Shared bus, central aggregator, and fast peer broadcast ($K \leq 4$) live in the M-saturation / C-collapse regime; independent and slow peer ($K \geq 16$) live in the M-deficit / C-saturation regime. The interior minimum at K=8 is robust in location but shallow in margin (Appendix F.3); the censoring-aware contrast is in the text.

location is robust but shallow — its margin over the adjacent fast cadences (K=1, 4) is not resolved under cell-level resampling (paired-bootstrap CIs in Appendices F.1 and F.3). Both statements condition on success. Under a censoring-aware expected time that charges failed runs the full step limit, the fast flank reverses and the independent baseline is marginally best (cell-paired $\Delta(K{=}8 - \text{independent}) = +0.32$, CI $[+0.05, +0.63]$; Appendix F.3): at fast cadence the completion decline reduces the success rate by more than the materialization gain increases it, so the interior optimum is a statement about conditional speed, not unconditional efficiency — and closing the unconditional gap is what the reservation protocol of open problem (3) achieves (Appendix F.10). Scaling to N=12 *sharpens* the signature: the peak Pearson rises to -0.70 and migrates outward to $K = 8$, consistent with stronger occupancy coupling at higher N delaying the cost-regime transition (Appendix E, Figure 7).

Provenance stratification: the mechanism behind the shift. Annotating each $M^\star$ lock with the *provenance* of the evidence behind it makes the bottleneck shift mechanistic rather than correlational. Let φ be the foreign share of the evidence mass supporting the locked candidate, computed from the same trust/support weights the merge itself used and frozen at lock time, and call a lock a *warp* ($W^\star$) when $\varphi \geq 0.8$: the agent is committing to a target it knows essentially only from peers. On a dedicated peer sweep over the scarcity cells (random layouts, 20 seeds, $K \in \{1, 2, 4, 8, 16\}$) the warp share $P(W^\star|M^\star)$ falls from 0.55 at K=1 to 0.11 at K=8 while the own-evidence stratum completes at a flat 0.26–0.34 and the warp stratum completes below 0.05 throughout: *the C-collapse of fast sharing is concentrated in the warp stratum*. What cadence controls, mechanically, is how much of the collective's materialisation runs on foreign evidence — and that is the stratum whose completions are lost to occupancy contention. The stratification is not an artefact of the 0.8 cutoff: recomputed from the raw per-lock φ values (23,237 locks) at thresholds $\varphi \in \{0.5, 0.6, 0.7, 0.8, 0.9, 0.95\}$, the foreign stratum completes at $\leq$0.03 while the own stratum stays at 0.26–0.50, at every threshold and cadence. The two coupling channels also separate interventionally with arms already in the design: the independent arm removes peer evidence while leaving target contention in place, and the reservation protocol of open problem (3) (Appendix F.10) keeps peer evidence while managing contention — under it, unconditional success at fast cadence exceeds the K=8 baseline (0.614–0.620 vs. 0.583; paired improvement over the same cells without the mechanism $\Delta = +0.037$, cluster-bootstrap 95% CI $[+0.018, +0.058]$ over 18 matched design cells), identifying occupancy contention rather than evidence quality as the cost channel of fast sharing. The same stratification reproduces on the continuous VMAS bridge ($0.72 \to 0.00$ over $K{=}1/4/16$) and on an LLM text substrate ($0.42 \to 0.08$ over $K{=}2/8$), with the curves nearly coinciding across substrates. A full provenance-conditioned analysis (counterfactual attribution by masked replay, a closed-form staleness horizon with a pre-specified $6/6$-exact hold-out, and a decentralized repair protocol) appears in a companion manuscript; here the stratification serves as the mechanistic reading of Empirical Claim 1.

What we do not claim about Φ. The conditional product $\Phi = P(M^\star|R^\star) \cdot P(C^\star|M^\star)$ does *not* have an interior peak in the tested range (mean-of-products estimator increases monotonically from 0.569 at $K{=}4$ to 0.603 at $K{=}64$, converging to the independent-agent boundary 0.605). The slope-level statement of Empirical Claim 1 is on the conditional rates themselves, not on Φ; we make no formal claim about the shape of $\Phi(K)$. Jensen-gap detail and POM–MOP comparison in Appendix F.1.

Learned-communication baseline: stage events are structurally degenerate even at competitive success. A CommNet-style policy (Sukhbaatar et al., 2016) trained with PPO (Schulman et al., 2017) on the same scenario ($N{=}3$, $T{=}2$) reaches 0.667 held-out success ($n{=}100$ seeds) — matching the symbolic peer architecture at $K{=}8$ (0.65). Despite this, the Q/R/M/C profile remains structurally degenerate: $Q^\star{=}1.00$ saturated by construction; $R^\star{=}0.997$, $M^\star{=}0.997$ collapsed together ($P(M^\star|R^\star){=}1.00$). This is the verified structural claim: a real-valued comm vector is never informatively empty (Q saturates) and lacks an explicit target-lock interface (R/M merge), regardless of training budget. The multi-agent counterpart of the single-agent learned-BC analysis (§3.3). Full architecture, PPO hyperparameters, and side-by-side REINFORCE vs PPO comparison in Appendix H.

LLM model sweep: the logger localizes a real failure in language agents. Table 4 runs the same Q/R/M/C event contract on four LLM agents over two independent backends: qwen3:4b and llama3.1 via local Ollama (8 episodes/cell), and Qwen2.5-3B/7B-Instruct via HuggingFace transformers on SLURM GPU nodes (25 episodes/cell). The decomposition separates *two different failure modes on two different stages*. **Budget → C:** below the task's step requirement, three models (qwen3:4b, llama3.1, Qwen2.5-7B) show the identical signature $Q^\star{=}R^\star{=}M^\star{=}1.00$, $C^\star{=}0.00$ — a completion-limited failure that replicates across backends, model families, and hardware. **Capability → R:** Qwen2.5-3B fails at *every* budget with $Q^\star{=}1.00$ but $R^\star{=}0.00$: it declares the target, emits well-formed actions (verified against the response cache, so not a parsing artifact), yet wanders without ever storing the evidence retrieval needs — an exploration/grounding failure surfacing at the R stage, exactly the "absence of evidence" regime of §3.5. An opaque scalar would report both models as 0.00; the stage view shows they fail for different reasons at different stages. (Instrumentation note: a naive qwen3:4b/Ollama chat-template call initially returned empty `response` strings because the model spent the short structured-output budget in a separate thinking field; we use raw structured completion for qwen3:4b and a first-line parser, recorded for reproducibility.)

The multi-agent phenomenology reproduces on a live LLM collective. Beyond the single-agent budget study, the same event contract instruments a collective of $N{=}3$ LLM agents (llama3.1) sharing symbolic place memory by peer broadcast under scarcity ($M{=}2$ takeable targets, four layouts, $K \in \{2, 8\}$), with all predictions specified before the runs. The cadence phenomenology of §4.5 carries over to the language substrate: a paired comparison against the no-communication baseline on identical layouts shows raw sharing *lowering* success from $2/3$ to $1/3$ per episode — the occupancy-racing C-collapse, now produced by agents whose queries, tag extraction, and commitments are LLM-made — and target collisions appear in $7/8$ shared-memory episodes. Attaching the reservation-and-rollback repair protocol (the constructive answer to open problem 3 of Appendix F.10) recovers part of the loss ($0.333 \to 0.417$ at both cadences, rollback latency again $\approx K$). Two interface findings transfer to any LLM-over-symbolic-memory design: LLM query formers free-generate retrieval tags outside the memory's vocabulary (requiring tolerant retrieval gates), and small local models are reliable at semantic commitment but not at coordinate arithmetic or systematic exploration (requiring layered execution). Full protocol and provenance analysis in the companion manuscript.

Scope: these are controlled studies on local models; scaling the same contract to full Voyager/MemGPT-style agents on richer tasks with multiple failure modes is developed next (external-framework validation) and in Appendix F.12.

Table 4: LLM TextNav step-budget sweep across four models and two independent backends (local Ollama, n=8/cell; HuggingFace transformers on a SLURM GPU cluster, n=25/cell; temperature 0.0; the task needs $\sim$10 low-level steps). Two distinct failure modes are localized to two different stages. **Budget → C:** at step limit 6, qwen3:4b, llama3.1 and Qwen2.5-7B all show $Q^\star$=$R^\star$=$M^\star$=1.00 with $C^\star$=0.00 — the completion-limited failure replicates across backends, models, and hardware. **Capability → R:** Qwen2.5-3B fails at *every* budget ($\{6, 8, 12, 25\}$) with $Q^\star$=1.00 but $R^\star$=0.00: it declares the target yet never accumulates the evidence to retrieve it (its outputs are well-formed action strings — verified in the response cache — so this is an exploration/grounding failure feeding retrieval, not a parsing artifact). Artifacts: `tmp/llm_steplimit_*`, `tmp/cluster/textnav*`.

Model	Backend	Step limit	Success	$Q^\star$	$R^\star$	$M^\star$	$C^\star$
qwen3:4b	ollama	6	0.00	1.00	1.00	1.00	**0.00**
llama3.1	ollama	6	0.00	1.00	1.00	1.00	**0.00**
Qwen2.5-7B	hf	6	0.00	1.00	1.00	1.00	**0.00**
qwen3:4b	ollama	12	1.00	1.00	1.00	1.00	1.00
llama3.1	ollama	12	1.00	1.00	1.00	1.00	1.00
Qwen2.5-7B	hf	12	1.00	1.00	1.00	1.00	1.00
Qwen2.5-3B	hf	6–25	0.00	1.00	**0.00**	0.00	0.00

Multi-agent oracle interventions close the diagnose→intervene→verify loop. On the scarcity cases, mean t_{succ} at small/intermediate cadence ($K \in \{1, 2, 4\}$) drops from $\sim$ 10–11 in baseline to $\sim$ 7.2 under either oracle R or oracle M, and further to $\sim$ 6.1 under oracle C, which additionally removes post-lock travel; at slow cadence (K=16) the baseline is already within $\sim$ 1.2 of the R/M oracle floor (Table 19 in Appendix F.13). Crucially, oracle C gives the largest time reduction but the smallest success-rate gain (Appendix F.14): in scarcity, completion costs time while retrieval and materialisation decide success. The cadence-induced excess time at small/intermediate K is attributable to the M↔C transition; at large K the bottleneck has moved into solo exploration.

The multi-agent block thus delivers its half of the argument. The cadence K is, mechanically, the rate at which private memories consolidate into a collective view; the data show that this rate has a non-trivial interior optimum ($K = 8$ on t_{succ}), that its cost channel is occupancy coupling, and that both effects are visible stage-by-stage on the same Q/R/M/C chain used for single agents. Consolidation is not only the single-agent limit; in a collective it is a tunable, measurable design axis.

5 The contract on third-party agent stacks

This section instantiates the last row of Table 1. We further evaluate the protocol on systems built by others. The contract has two layers: an *invariant core* — four event roles (a memory query is issued; returned evidence satisfies the task; the agent commits to a target; the task completes) — and a *system-specific adapter* that maps logged actions to those roles. We instrument three third-party agent stacks — a function-calling loop on the OpenAI SDK, LangGraph's prebuilt ReAct agent, and AutoGen's AssistantAgent/UserProxy pair, all driving the same local model (llama3.1, temperature 0) — with one tool-trace adapter shared by all three: an episodic household memory is exposed through `recall`/`goto`/`take` tools, and $Q^\star$ fires on querying memory, $R^\star$ on *attribution* (a returned record names the item at its true place), $M^\star$ on a commitment (a `goto` to the true place; first-commitment accuracy M_1 is logged as an auxiliary lock-quality diagnostic), and $C^\star$ on task completion within a tool budget. A tool trace exposes no lock chain, so these are *marginal event frequencies* $f(\cdot)$ in the sense of Appendix C: an agent can commit to the true place without a supporting retrieval, the events are not nested, and no factorisation is claimed — the four frequencies serve as separate diagnostics and are never substituted into the chain-rule product. What varies across system classes is the adapter, not the event roles. Three variants engineer one failure each — a consolidation gap (the location fact was never written to memory), a stale-distractor ambiguity, a tool budget below the task's requirement — plus a clean control; all predictions were specified before the runs, and the two revisions of the adapter (an attribution leak; an over-tightened $M^\star$) are recorded in the registration file.

Table 5: Marginal Q/R/M/C event frequencies on three third-party agent frameworks (llama3.1, 20 episodes per cell, identical episodes/tools/budgets). Each entry is the fraction of episodes in which the event occurred: $Q^\star$ — the agent queried memory at least once; $R^\star$ — a returned record named the item at its true place; $M^\star$ — the agent committed (`goto`) to the true place, with or without retrieval support; $C^\star$ — `take` succeeded within the tool budget; M_1: first `goto` targets the true place; stale$_1$: first `goto` targets the stale distractor. The events are not nested and no product is claimed (Appendix C): under the consolidation gap, budgeted blind search over the four places sometimes commits to and reaches the true place, hence $M^\star, C^\star > 0$ at $R^\star$=0. Structural failures localize identically everywhere: the consolidation gap is $R^\star$=0.00 and the budget failure is $C^\star$=0.00 at $Q^\star$=$R^\star$=1, for all three stacks. On the *clean* control the frameworks spread from 0.50 to 0.75 end-to-end — and the decomposition attributes the spread (see text).

Framework	Variant	$f(Q^\star)$	$f(R^\star)$	$f(M^\star)$	$f(C^\star)$	$f(M_1)$	$f(\text{stale}_1)$
OpenAI SDK	control	1.00	1.00	0.85	0.75	0.25	—
	consolidation gap	1.00	**0.00**	0.50	0.40	0.25	—
	stale ambiguity	1.00	1.00	0.90	0.30	0.25	0.20
	tight budget	1.00	1.00	0.25	**0.00**	0.25	—
LangGraph	control	1.00	1.00	0.70	0.50	0.50	—
	consolidation gap	1.00	**0.00**	0.20	0.15	0.10	—
	stale ambiguity	1.00	1.00	0.65	0.60	0.50	0.35
	tight budget	1.00	1.00	0.10	**0.00**	0.10	—
AutoGen	control	1.00	1.00	0.85	0.55	0.45	—
	consolidation gap	1.00	**0.00**	0.30	0.20	0.15	—
	stale ambiguity	1.00	1.00	0.50	0.45	0.35	0.40
	tight budget	1.00	1.00	0.15	**0.00**	0.15	—

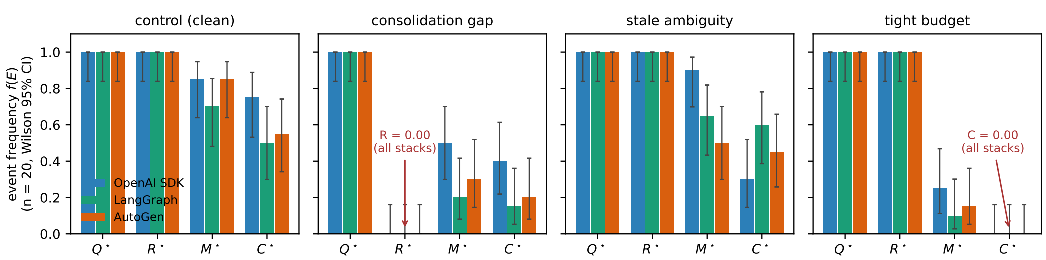

Figure 5: Stage profiles of three third-party agent stacks on the four engineered variants (llama3.1, n=20 per cell). Query and retrieval saturate identically everywhere; the two structural failures produce exact zero columns on every stack ($R^\star$=0 under the consolidation gap, $C^\star$=0 under the tight budget); the behavioral spread between frameworks lives entirely in the M and C stages.

Three findings (Table 5, Figure 5). **(1) Structural localizations are exact and framework-independent.** When the memory lacks the fact, $R^\star$=0.00 on every stack; when the budget cannot fit completion, $C^\star$=0.00 at $Q^\star$=$R^\star$=1 on every stack (20/20 episodes each). The contract needs no per-framework adaptation. **(2) The clean control separates the frameworks, and the decomposition says how.** End-to-end success on the SAME clean episodes ranges 0.50 (LangGraph) to 0.75 (OpenAI SDK). The stage profiles attribute the spread to a failure mode common to all three stacks combined with framework-specific recovery: first-commitment accuracy is low everywhere (M_1=0.25–0.50 — the agent's first `goto` frequently ignores its own just-retrieved evidence), episode-level materialization recovers to 0.70–0.85 through re-query loops, and the frameworks then differ in how much of that recovery survives to completion ($C^\star/M^\star$ from 0.88 for the OpenAI SDK loop down to 0.65 for AutoGen). **(3) The stale distractor pulls, measurably and unevenly.** The outdated record captures the first commitment in 0.20–0.40 of ambiguous episodes (most strongly for AutoGen), and its completion cost is framework-specific: the OpenAI SDK loop pays hardest ($C^\star$: 0.75 → 0.30) while LangGraph and AutoGen absorb the detour (0.50 → 0.60; 0.55 → 0.45) — the LLM-memory analogue of the staleness gating that the collective experiments showed to be decisive. Honestly failed registrations: the behavioral predictions that assumed a clean control (all stages ≥ 0.75; a uniform ambiguity-induced completion drop) broke *because the frameworks* *themselves fail a clean memory task 25–50% of the time and pay for ambiguity in different places* — which is finding (2)–(3), not noise; all revisions are recorded as failed pre-specifications with mechanisms.

Model robustness: the structure transfers, the behaviour is legible. A prediction specified before the runs — the structural localizations transfer to a second model on all frameworks, while behavioural rates are model-dependent and only reported — was confirmed exactly by repeating all 240 episodes with qwen3:4b (Table 7): $R^{\star}$=0.00 under the consolidation gap and $C^{\star}$=0.00 under the tight budget, 20/20 on every stack, for both models. The behavioral contrast is a diagnostic result: qwen3:4b completes every clean episode ($C^{\star}$=1.00 across stacks, vs. 0.50–0.75 for llama3.1) because its *first-commitment accuracy* is high (M_1=0.90–0.95 vs. 0.25–0.50) and its susceptibility to the stale distractor low (0.05–0.30) — except under AutoGen, where qwen's M_1 collapses to 0.40 while remaining ≥ 0.90 on the other stacks. The stage decomposition thus separates three influences a leaderboard conflates: the task's structural limits (model- and framework-independent), the model's commitment quality, and the framework's prompt/loop shaping that can degrade it.

Comparison with diagnostic baselines. On the blinded benchmark below we additionally evaluate four alternative diagnostics of the same logs, each receiving the identical anonymized cell aggregates and calibration control, with decision rules fixed before scoring and thresholds matched to the Q/R/M/C rule (`experiments/qrmc_external/diagnostic_baselines.py`; specification in `tmp/qrmc_external/baselines_spec.json`). Over 63 scored cells (10 faults + held-out control, three stacks, two models): a success-only reading names no stage (0.095); an AgentBoard-style progress scale (first milestone drop) reaches 0.730 (macro-F1 0.698); a RAGChecker-style two-stage retrieval/execution split reaches 0.651 exact / 0.873 side-level (macro-F1 0.450), its deficit concentrated where commitment must be separated from execution; a process-conformance first-deviation rule using the order-sensitive M_1 reaches 0.778 (macro-F1 0.755), its entire deficit being the budget-exhaustion branch of the conditional reading; a multinomial logistic regression on the same aggregates reaches 0.444 on unseen fault types with a false-positive rate of 1.00 on held-out controls (leave-one-model -out 0.794, leave-one-framework-out 0.825) and requires fault labels that the rule-based methods do not. Q/R/M/C attains 0.873 (macro-F1 0.835; bootstrap CI over fault types $[0.71, 0.98]$) and outperforms all exact-stage alternatives under paired McNemar tests, with all comparisons significant after Holm correction (largest adjusted $p = 0.031$; discordant cells from 6:0 to 50:1). Of the 66 cells, three (the prompt-only fault under llama3.1) failed the pre-specified manipulation check and are excluded for every method alike, leaving 63. A follow-up repair study (llama3.1, 2,640 episodes; pre-specified menu of stage-level repairs, each diagnostic selecting the repair for its predicted stage) gives Q/R/M/C the highest completion gain and lowest regret against the best available arm (0.293/0.061 vs. an oracle-best gain of 0.354; two-stage 0.280/0.074; conformance 0.250/0.104; progress 0.219/0.135; random 0.113/0.240). Paired bootstrap over fault–framework cells resolves the gain advantage over the progress baseline ($+0.074$ $[+0.007, +0.157]$) but not over the two-stage baseline ($+0.013$ $[-0.009, +0.039]$), which also selects the single best arm slightly more often (0.704 vs. 0.630) because its coarse diagnoses map to repairs that frequently coincide with the best available arm; the four-stage advantage is established at exact localization, while on repair utility the protocol at least matches the strongest coarse baseline. Two pre-specified expectations failed: the matched repair for silent corruption recovers only 0.22–0.44 of the control gap (consolidation without decoy removal converts a retrieval fault into an ambiguity fault), and the budget repair on the clean OpenAI SDK control lowered completion by 0.15.

Blinded fault-localization meta-evaluation (pre-specified). The engineered variants verify localization of failures whose stage is known by construction; the reviewer-grade test treats the protocol as a *classifier*. We injected ten fault types — two to three per stage, structural and behavioural (Table 6) — and fixed a first-failing-link decision rule, with all thresholds, *before* any run (`tmp/qrmc_external/registered_meta.json`; rule and thresholds in `experiments/qrmc_external/run_meta_evaluation.py`). The rule receives only anonymised marginal profiles plus a labelled control cell for calibration; a held-out control (seeds 20–39) enters the blind set with ground truth "none", measuring the false-positive rate; a manipulation check excludes faults that do not bind (the prompt-only fault moved qwen but not llama). Stage-level accuracy: 26/30 scored cells for llama3.1 (8/10, 8/10, 10/10 per stack) and 29/33 for qwen3:4b (9/11, 10/11, 10/11); the five structural faults localize 30/30 across both models and all stacks; held-out controls read "none" in 5/6 cells (one false positive at n=20); cross-framework label agreement is 7/11 (llama) and $10/11$ (qwen). The pre-specified model-contrast prediction was confirmed in its named direction: M-family faults hide behind llama's weak baseline commitment (M_1=0.25–0.50) and become visible on qwen's clean baseline — the unmarked duplicate drops M_1 by $0.90/0.80/0.20$ across stacks — while its accuracy clause held on two of three stacks (AutoGen: $10/11$ vs. llama's $10/10$, a one-cell difference). One pre-specified risk materialised instructively: the *marked* stale distractor changes llama3.1's behaviour but reads "none" on qwen3:4b, which takes the staleness marker into account — the protocol measures the behavioural effect of a fault, not its presence, and a fault a model neutralizes is reported as no failure.

Table 6: Blinded fault localization: injected fault, ground-truth stage, and correct-localization count over the three stacks, per model. "Neutralized": the fault does not change qwen's behaviour (the staleness marker is heeded), so the blind rule reads "none".

Injected fault	Stage	llama3.1	qwen3:4b
recall tool absent	Q	3/3	3/3
prompt: memory unreliable	Q	non-binding	3/3
fact never consolidated	R	3/3	3/3
silent corruption	R	3/3	3/3
flaky recall (p=0.7)	R	3/3	3/3
stale distractor (marked)	M	1/3	0/3 (neutralized)
duplicate (unmarked)	M	2/3	3/3
budget below requirement	C	3/3	3/3
flaky goto (p=0.5)	C	2/3	3/3
take always fails	C	3/3	3/3
held-out control	none	3/3	2/3

Table 7: Second-model replication (qwen3:4b, 20 episodes per cell, identical episodes/tools/budgets; thinking disabled request-level for budget parity, recorded in the pre-run specification). Entries are marginal event frequencies $f(E)$ as in Table 5 — not chain-rule links. Structural zeros transfer exactly; behavioural rates differ from llama3.1 (Table 5) in the direction the M_1 diagnostic explains.

Framework	Variant	$f(Q^\star)$	$f(R^\star)$	$f(M^\star)$	$f(C^\star)$	$f(M_1)$	$f(\text{stale}_1)$
OpenAI SDK	control	1.00	1.00	1.00	1.00	0.95	—
	consolidation gap	1.00	**0.00**	0.45	0.45	0.20	—
	stale ambiguity	1.00	1.00	1.00	0.95	0.90	0.05
	tight budget	1.00	1.00	0.30	**0.00**	0.30	—
LangGraph	control	1.00	1.00	1.00	1.00	0.90	—
	consolidation gap	1.00	**0.00**	0.60	0.60	0.30	—
	stale ambiguity	1.00	1.00	0.95	0.95	0.75	0.20
	tight budget	1.00	1.00	0.15	**0.00**	0.15	—
AutoGen	control	1.00	1.00	1.00	1.00	0.40	—
	consolidation gap	1.00	**0.00**	0.55	0.50	0.30	—
	stale ambiguity	1.00	1.00	1.00	0.90	0.40	0.30
	tight budget	1.00	1.00	0.15	**0.00**	0.15	—

Scope: three stacks at n=20 per cell on two local models; Letta/CrewAI (Python $\geq$3.10) and API-scale models are the natural extension.

6 Limitations and future work

Portability across substrates: slope conditions hold on MiniGrid. The slope conditions of Empirical Claim 1 also hold on a standard-MiniGrid substrate. A FourRooms-based multi-agent wrapper (built from `minigrid.core.grid.Grid` + Wall/Floor/Lava primitives, identical operational $Q^\star, R^\star, M^\star, C^\star$ definitions; `experiments/minigrid_multiagent_wrapper/`) was used to run a 100-run portability sweep ($K \in \{1, 4, 8, 16, 64\}$, 20 seeds, N=3, T=2): Spearman $P(M^\star|R^\star)$ vs K is -0.50 and Spearman $P(C^\star|M^\star)$ vs K is $+0.50$ (moderate rank correlations; this is a small 100-run sweep, so we report the effect sizes and do not lean on significance). The interior t_{succ}

minimum does not separate at this minimal sweep size (5 cadences, 20 seeds, one hazard density); a denser sweep is the natural next step. The two-substrate seam is therefore closed at the slope-level claim.

Empirical Claim 1 is slope-level on conditional rates only. The two slope conditions hold as moderate, cluster-robust effect sizes (Appendix F.3); we make no formal claim on the shape of the conditional product Φ (§4.3). A future statement directly in terms of the time-cost surface, or a derivation under a generative model (Poisson-renewal memory, birth-death occupancy), would convert the empirical claim into a theorem.

Estimability under hidden state. With unlogged in-episode query rewrites or peer side-channels, Definitions 1 and 2 are projections onto logged traces, not full identification.

Shared coordinate frame. All multi-agent results assume agents interpret places in a common coordinate frame: cross-agent place identity is coordinate proximity, with semantic profiles as a tie-breaker inside a spatial radius. The place-correspondence problem — whether two agents' records denote the same place at all — is therefore bracketed here, not solved. A companion manuscript removes the assumption constructively (fingerprint-based place identity with consensus frame recovery up to translation and $90°$ rotation) and shows the Q/R/M/C measurements, including the cadence slopes and the staleness horizon, unchanged on top of the recovered frames; the harder variants (continuous $SE(2)$, appearance noise, symbol alignment) remain open.

The evaluator's correspondence oracle. A deeper form of the same dependence survives every representational relaxation, and we state it explicitly. The protocol unifies *observation points*, not internal representations — but scoring the two middle stages presupposes an *entity correspondence* between memory content and world referents: $R^\star$ asks whether a returned candidate denotes the true target, $M^\star$ whether the lock does. In every study here that oracle is supplied by an instrumented environment (ground-truth cells on the grids; the task's true location in the tool-trace contract). The dependence is graded, which is what makes the protocol degrade gracefully rather than break: $Q^\star$ requires no correspondence at all (did the agent query its memory), $C^\star$ is behaviourally observable (did the task complete), while $R^\star$ and $M^\star$ — the diagnostic payload — are undefined without it. The severity of the requirement depends on the population under evaluation: interchangeable agents (a shared perception interface, possibly different weights) situated in a single environment inherit the correspondence from the shared substrate — the setting of every study in this paper and the common case in multi-agent evaluation. It becomes a genuinely open problem for heterogeneous agents in open worlds, where deploying the protocol requires either an instrumented task or a correspondence mechanism with its own error model; the companion's meaning-based place identity is a candidate for grounding the *evaluator's* correspondence, not only the agents', and we leave that construction open.

Learned baselines. The CommNet PPO baseline reaches 0.667 success on the scarcity scenario — competitive with symbolic peer at $K{=}8$ (0.65) — and confirms Q-saturation ($Q^\star{=}1.00$) and R/M collapse ($P(M^\star|R^\star){=}1.00$) at competitive success. The structural prediction now holds verified, not asserted; ablating on architectures with explicit symbolic target-lock channels is left for future work.

Scale. Multi-agent sweep bounded at $N{=}12$ (Appendix E; signature sharpens, peak Pearson -0.70 at $K{=}8$, interior t_{succ} minimum 4.77). Denser scaling at $N{\geq}16$ is future work. **Continuous substrate.** A VMAS (Bettini et al., 2022) validation reproduces *both* slope signs of Empirical Claim 1 across three scarcity cells $\times$ 40 seeds with block-bootstrap CIs excluding zero (Spearman -0.999 $[-1.000, -0.975]$ and $+0.995$ $[+0.900, +1.000]$ over $K{\leq}16$; Appendix F.7), reusing the symbolic memory through a discretisation bridge. The shift is therefore not an artefact of grid cells; what remains open is fully continuous *memory* (the bridge discretises it) and visually rich 3D substrates.

7 Broader impacts

The paper proposes diagnostic evaluation for memory-based agents in simulated navigation. Its positive impact is methodological: making memory-based failures more measurable across architectures, and making the cadence axis of distributed multi-agent memory tunable on an empirically grounded criterion. Potential negative impacts are indirect and would arise if similar diagnostics were used to improve agents in safety-critical settings (drone swarms, autonomous vehicles, distributed sensor networks) without adequate safeguards. The current work is benchmark-level infrastructure, not a deployment-ready system.

8 Conclusion

We proposed Q/R/M/C, a four-stage decomposition of memory-based navigation whose factors are estimable from standard trajectory logs, and used it as a diagnostic instrument across symbolic, learned, LLM-driven, single-agent, and multi-agent traces. The main result is not a new agent architecture: it is a reproducible logging and estimation protocol that separates query formation, retrieval satisfaction, target materialization, and post-retrieval completion. Across the experiments, this protocol localizes failures that success-only evaluation cannot, exposes degenerate observability in implicit learned policies, and makes memory-sharing cadence measurable as a stage-level design variable. The resulting rates are directly comparable across systems that expose a query/lock interface; for opaque learned policies the protocol measures stage observability itself.

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A Formal proofs

A.1 Argument for Definition 1 (single-agent estimability)

For each episode $i \in \{1, \dots, n\}$ let $Q_i^\star, R_i^\star, M_i^\star, C_i^\star$ denote the nested episode-level indicators. By construction these are measurable functions of the runtime trajectory log and satisfy $C_i^\star \Rightarrow M_i^\star \Rightarrow R_i^\star \Rightarrow Q_i^\star$. The empirical estimators

$$\hat{p}_Q = \tfrac{1}{n} \sum_i Q_i^\star, \quad \hat{p}_{R|Q} = \tfrac{\sum_i R_i^\star}{\sum_i Q_i^\star}, \quad \hat{p}_{M|R} = \tfrac{\sum_i M_i^\star}{\sum_i R_i^\star}, \quad \hat{p}_{C|M} = \tfrac{\sum_i C_i^\star}{\sum_i M_i^\star}$$

are consistent by the chain rule plus the conditional stage-Markov property, and the empirical product converges to semantic-path completion. □

A.2 Argument for Definition 2 (multi-agent estimability)

Conditioning sets are enlarged by $\mathcal{M}_{-i}^{(K)}$ and $\mathcal{O}_{-i}$. Under the enlarged stage-Markov property, the proof of Definition 1 transfers verbatim and each enlarged conditional is a Bernoulli probability on a measurable event. The qualifications stated in §4.2 (“When Definition 2 may fail”) describe two specific situations in which the enlarged stage-Markov property is violated. □

A.3 Argument for Empirical Claim 1 (slope-level statement)

Empirical Claim 1 is an empirical claim, not a theorem. The argument below is the qualitative justification that motivates the two slope signs; the data are what verify them. We deliberately do not call this a proof.

Slope (a), under (F). As K decreases, $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise. Conditional on a successful retrieval, the probability of locking onto a reachable target is non-decreasing in ν. Therefore $\partial P(M^\star | R^\star; K)/\partial K^{-1} > 0$.

Slope (b), under (O). As K decreases, peers receive updated memory simultaneously and converge on the same closest target; the marginal occupancy $\Pr[\text{target} \in \mathcal{O}_{-i}]$ rises. Therefore $\partial P(C^\star|M^\star;K)/\partial K^{-1} < 0$.

Note on the conditional product Φ. Empirical Claim 1 makes no statement about the shape of $\Phi(K) = P(M^\star|R^\star;K)\cdot P(C^\star|M^\star;K)$. Under (F) and (O) the slope of Φ in K^{-1} is the sum of two opposing terms, and whether it changes sign depends on quantitative magnitudes that are not pinned down by (F) and (O) alone. In our data Φ is monotone in K over $\{4, 8, 16, 32, 48, 64\}$ and converges to the independent-agent boundary $\Phi_\infty = 0.605$. The practically relevant efficiency surface — mean time-to-success — does have an interior minimum at $K = 8$; see §4.5. □

A.4 The query–retrieval Galois connection: statement, argument, scope

Fact (standard). Let G and Σ be sets and $I \subseteq G \times \Sigma$ a relation. For $B \subseteq \Sigma$ define $\operatorname{ext}(B) = \{p \in G : \forall \tau \in B,\ (p, \tau) \in I\}$, and for $A \subseteq G$ define $\operatorname{int}(A) = \{\tau \in \Sigma : \forall p \in A,\ (p, \tau) \in I\}$. Then for all $A \subseteq G$, $B \subseteq \Sigma$:

$$A \subseteq \operatorname{ext}(B) \iff B \subseteq \operatorname{int}(A);$$

both maps are antitone, both composites $\operatorname{ext} \circ \operatorname{int}$ and $\operatorname{int} \circ \operatorname{ext}$ are closure operators, and the closed pairs (A, B) with $A = \operatorname{ext}(B)$ and $B = \operatorname{int}(A)$, ordered by extent inclusion, form a complete lattice — the concept lattice of the context (G, Σ, I) (Ganter and Wille, 1999).

Argument. $A \subseteq \operatorname{ext}(B)$ unfolds to $\forall p \in A\ \forall \tau \in B : (p, \tau) \in I$, a condition symmetric in A and B, which is exactly $B \subseteq \operatorname{int}(A)$. Antitonicity and the closure properties follow by the standard one-line arguments (Ganter and Wille, 1999, Ch. 1). Viewing the power sets as thin categories (one with reversed order), the displayed equivalence says precisely that int and ext form an adjoint pair; a Galois connection is an adjunction between posets (Mac Lane, 1998, Ch. IV). □

Instantiation and scope. In our memory, take G to be the place records, Σ the tag alphabet, and $I_\theta = \{(p, \tau) : \text{the per-place tag table assigns } \tau \text{ to } p \text{ with confidence } \geq \theta\}$ (Appendix D); the intent of a query is $B = q^{\text{req}}$ and R-availability is $\operatorname{ext}(B) \neq \emptyset$. Three deliberate scope restrictions. (i) *Graded scores.* The deployed retrieval score is graded (the weighted log-score over required, preferred, and penalty tags of Appendix D), so the crisp connection above formalizes the thresholded required-tag skeleton of candidacy; the graded analogue is the theory of fuzzy Galois connections (Bělohlávek, 1999) and we do not use it here. (ii) *Non-monotone updates.* Dirichlet–multinomial updates are not literally monotone in I_θ: contrary evidence can remove an incidence. On the reported benchmarks the binding failure is absence of evidence rather than contradiction (91% empty extents, 96.9% precision when non-empty, §3.5), so the monotone-growth reading is the empirically relevant regime, and we flag the exception rather than assume it away. (iii) *No structure for M or C.* Materialization is a satisficing choice function on non-empty extents and Completion is a control problem under occupancy; neither carries the order structure, and we attach no categorical claims to them. The reading earns its place by naming the failure mode exactly (a queried intent whose extent is empty lies outside the realized concept lattice), by giving consolidation a precise object (growth of the context, hence of every extent), and by locating the cadence axis structurally: memory coupling $\mathcal{M}_{-i}^{(K)}$ merges peer contexts at rate K^{-1}, while occupancy $\mathcal{O}_{-i}$ is extra-contextual — which is the structural content behind the two slope signs of Empirical Claim 1.

B An options / hierarchical-MDP reading of the stages

Why a deterministic strategy: bounded and ecological rationality

A natural objection is that the semantic stack is a fixed, deterministic strategy rather than a learned policy. We treat the determinism as principled and name it: the pipeline is a satisficing agent in Simon's sense (Simon, 1955, 1956). Candidacy is threshold-gated instead of exhaustively scored, the target commitment is made once instead of re-optimized at every tick, and the intent is a few tags instead of the full state. Bounded rationality is also what makes the four failure loci observable: each stage is a distinct resource bound (attention, memory, deliberation, control), and Q/R/M/C is the audit trail of where the bound binds. The learned baselines make the converse point empirically: policies without an explicit query/lock interface collapse the stages even at competitive success

(BC in §3.3; CommNet PPO in §4.5), so observability is traded away exactly where end-to-end optimization is traded in. That the *same* heuristic is strong on water and rest, retrieval-starved on hazard recovery, and downstream-limited on goal-region (Figure 1b) is the ecological-rationality point that heuristic success is a property of the heuristic–environment pair (Simon, 1956; Gigerenzer and Goldstein, 1996); the framework is the instrument that locates the mismatch. The same lens reads the Φ-versus-t_{succ} split of §4.5: the chain probability Φ ignores the cost of traversing the chain, so under a resource-rational reading (Lieder and Griffiths, 2020) an optimum should be sought on a cost-bearing metric; that is where it appears ($K{=}8$ on mean time-to-success) and not on Φ.

This appendix also develops the dynamic reading pointed to from §2.3. It introduces no new algorithm: it names the object the pipeline already computes and makes the Φ-versus-t_{succ} split a property of that object. Where the order-theoretic reading (Appendix A.4) fixes the *static* content of the stages, this one addresses time and cost.

Q/R/M/C as an audit of an option. Each query instantiates one temporally extended action—an option $o = \langle \mathcal{I}, \pi, \beta \rangle$ in the sense of Sutton et al. (1999): an initiation set $\mathcal{I}$, an internal policy π, and a termination condition β. Standard trajectory-level analysis treats an option as a black box scored by its return. The Q/R/M/C decomposition is instead an audit of the option's *internal* structure, and it lines up edge-for-edge with the factorization $Q^\star \to R^\star \to M^\star \to C^\star$:

$Q^\star \to R^\star$ **(initiation).** The option "reach a place satisfying q" is well-initiated only when its initiation set is non-empty. In the vocabulary of §2.3 this is exactly $\mathrm{ext}(q^{\mathrm{req}}) \neq \emptyset$: an empty extent is not a bad choice of option but the absence of any admissible one, so R fails before a target is ever locked.

$R^\star \to M^\star$ **(policy over options).** Materialization is the high-level policy that commits to a single option out of the admissible set—the target lock $g^\star = \chi(\mathrm{ext}(q^{\mathrm{req}}))$ under the satisficing choice function χ of §2.3.

$M^\star \to C^\star$ **(execution to termination).** Completion is the roll-out of the internal policy π until the termination condition β fires (arrival at $g^\star$, or timeout). This is the only edge whose failure modes are extra-semantic: controller competence and, in a collective, occupancy $\mathcal{O}_{-i}$.

So Q/R/M/C is, on this reading, a stage-resolved instrument for the internal structure of a hierarchical-MDP option generator, rather than a bespoke logger for a single environment.

Why the optimum lives on t_{succ} and not on Φ: a semi-MDP reading. In the options / semi-MDP formalism, an option is evaluated by a return accumulated over a *random* execution duration τ,

$$R(s, o) = \mathbb{E}\Big[\textstyle\sum_{t=0}^{\tau-1} \gamma^t \, r_{t+1} \;\Big|\; s_0 = s,\, o\Big],$$

where τ is the number of low-level steps until β fires (Sutton et al., 1999). The chain probability $\Phi = P(M^\star|R^\star)\cdot P(C^\star|M^\star)$ records only *whether* the option-audit succeeds; it is by construction blind to τ, which is the cost of traversing it. The cost-bearing metric t_{succ} is τ. This is the structural reason—and not merely an empirical coincidence—that the resource-rational reading of §2.3 (Lieder and Griffiths, 2020) locates an interior optimum on t_{succ} ($K{=}8$) while Φ stays monotone: the two metrics measure different faces of the same option, success versus duration.

The bottleneck shift as multi-agent option collision. The dynamic reading makes the M$\leftrightarrow$C trade-off of Empirical Claim 1 physically concrete. Faster broadcast (smaller K) merges peer contexts $\mathcal{M}_{-i}^{(K)}$ more often, which can only enlarge each agent's extent and therefore raises the probability that a target is admissible and locked: $P(M^\star|R^\star)$ increases. But because agents share a spatial basis and a tag alphabet (§2.1), fresher shared memory also *synchronizes* the high-level policy: several agents commit to the same option over the same scarce target. Under occupancy $\mathcal{O}_{-i}$ this inflates the execution duration τ and depresses $P(C^\star|M^\star)$. Cadence K thus regulates the dispersion of the agents' option distribution: rare exchange desynchronizes option selection and shortens τ, giving the minimum of t_{succ} at an interior K. The bottleneck is therefore grounded in a competition between initiation-set growth (semantic, contextual) and option-collision cost (extra-semantic, occupancy), exactly the two edges the cadence protocol $\mathcal{C}_K$ acts on in Figure 2.

Belief–desire–intention gloss and scope. The high-level state audited above can be named in belief–desire–intention terms, which is the form used by the collective-memory companion: *belief* is the realized concept lattice of §2.3 (what is retrievable), *desire* is the state-conditioned predicate that selects the query q (e.g. a thirst variable switching the required tags to `water_source`), and *intention* is the locked option $g^\star$. The hierarchical state is then $h = (\,\text{lattice},\ \text{desire},\ g^\star,\ s\,)$ with s the low-level physical state, and the three failure loci separate as before: an empty extent is a belief/R failure, a non-empty extent with no stable lock is an M failure, and a lock unreachable under occupancy is a C failure. Two scope caveats. We claim no options-*learning* result: the pipeline is a fixed satisficing option generator (Simon, 1955) and Q/R/M/C audits it. And we attach the option structure only to the semantic path—$\mathcal{I}$ to $Q \to R$, (π, β) to $M \to C$—while the semi-MDP return is used as an explanatory lens for the Φ-versus-t_{succ} split, not fitted to the data. Neither this options reading nor the order-theoretic reading of Appendix A.4 is claimed as a contribution: both are standard constructions used as interpretive lenses on the measurement protocol, which is the paper's contribution. Their value is that each makes a testable statement on the logs—an empty extent localises the R failure, and the semi-MDP duration τ predicts that the optimum appears on t_{succ} and not on Φ.

C Operational vs. nested stage estimators

Operational diagnostic rates. The main benchmark logs per-query and per-episode diagnostic quantities directly from runtime traces (query formed, retrieval satisfied, target materialized, completion after retrieval). These are operational rates, not direct multiplicative factors.

Nested stage estimators for factorization. For the factorization, episode-level nested events from the first failure stage recorded in the runtime taxonomy are used: $Q^\star, R^\star, M^\star, C^\star$, nested by construction. The consistency audit in §3.2 reports $\hat{P}(Q^\star)$, $\hat{P}(R^\star|Q^\star)$, $\hat{P}(M^\star|R^\star)$, $\hat{P}(C^\star|M^\star)$, their product, and its gap to semantic-path completion.

C.1 Multi-agent operational definitions

For the multi-agent sweep, every metric is defined per agent and per tick, then aggregated. Let $\mathrm{Targets}_i(t)$ be the candidate set returned by the memory query of agent i at tick t, $\mathrm{Locked}_i(t)$ be the agent's planner-internal locked target, and $\mathcal{W}$ the ground-truth target set.

Per-tick events. $Q_i(t) = \mathbf{1}[\mathrm{Targets}_i(t) \neq \emptyset]$; $R_i(t) = \mathbf{1}[\exists w \in \mathcal{W} : \min_c \|c - w\| \leq \epsilon]$ ($\epsilon = 0.6$); $M_i(t) = \mathbf{1}[\mathrm{Locked}_i(t) \neq \emptyset \wedge \min_w \|\mathrm{Locked}_i(t) - w\| \leq \epsilon]$.

Episode-level starred events. $Q_i^\star = \bigvee_t Q_i(t)$; analogously for $R^\star, M^\star$; $C_i^\star = \mathbf{1}[\text{agent } i \text{ reached a target}]$.

Aggregation. Per-run rates are averaged over the N agents; per-configuration rates are averaged over the 40 seeds. Conditional rates $P(R^\star|Q^\star)$, $P(M^\star|R^\star)$, $P(C^\star|M^\star)$ are computed as ratios of episode-level counts within each configuration before averaging across configurations.

D Memory and task schema

D.1 Full notation table

Symbols are fixed throughout the paper.

Stages. Q, R, M, C — the four stages: **Q**uery formation, **R**etrieval, **M**aterialization, **C**ompletion after retrieval. They name both the stage and (in expressions like "the M link") the corresponding edge of the chain.

Events. $Q^\star, R^\star, M^\star, C^\star \in \{0, 1\}$ — episode-level starred events recording whether each stage succeeded (Appendix C). Nested by construction: $Q^\star{=}1 \Leftarrow R^\star{=}1 \Leftarrow M^\star{=}1 \Leftarrow C^\star{=}1$.

Probabilities. $P(\cdot)$ — population-level probability (averaged over the seed distribution and, in the multi-agent case, over agents). $\hat{P}(\cdot)$ — empirical-frequency estimator computed from logs.

Single-agent. q — query; $\mathcal{M}$ — memory state; Y — end-to-end task success (can exceed $C^\star$).

Multi-agent. N — number of agents in the population. T — number of targets in the environment (constraint $T \leq N$; "scarcity" means $N > T$). i — agent index. $\mathcal{M}_i$ — agent i's private memory. q_i — agent i's query.

Coupling. $\mathcal{C}_K$ — the peer-broadcast communication protocol with cadence $K \in \mathbb{N}$ (every K ticks each agent broadcasts a memory snapshot, §4.1). $\mathcal{M}_{-i}^{(K)}$ — the latest peer memory snapshots accessible to agent i under $\mathcal{C}_K$ (the *memory coupling* channel). $\mathcal{O}_{-i}$ — the set of targets already claimed by other agents by the time i acts (the *occupancy coupling* channel).

Slope variables. $\nu(\mathcal{M}_{-i}^{(K)})$ — a freshness functional on peer memory; non-decreasing pointwise as K decreases (assumption **(F)** of Empirical Claim 1). The occupancy slope is assumption **(O)**.

Summary statistics. $\Phi(K) := P(M^\star|R^\star;K) \cdot P(C^\star|M^\star;K)$ — the conditional product, the chain probability $P(M^\star \wedge C^\star \mid R^\star)$. POM (product-of-means) and MOP (mean-of-products) are two estimators of Φ; they differ by $\mathrm{Cov}(P(M^\star|R^\star), P(C^\star|M^\star))$, which is the Jensen gap (§4.5). t_{succ} — mean time-to-success in environment ticks.

D.2 Memory stack and task patterns

The memory stack stores symbolic place records (per-place tag confidences, counts, visitation, freshness, geometry), transition records (success, risk, cost statistics with fast/slow estimates), episodic traces (intent + state at each visit), and concept/relation records. At each step, the tokenizer emits (z_t, τ_t, e_t) from o_t. Per-place tag tables use a Dirichlet-multinomial update; transitions use exponentially smoothed statistics; episodic records attach the active intent and agent state. A planner query is answered by retrieving candidate places whose semantic profiles match the current task intent and state, via a weighted log-score $s(p \mid q_t, s_t) = \sum_{\tau \in q^{\mathrm{req}}} \log P(\tau|p) + \alpha \sum_{\tau \in q^{\mathrm{pref}}} \log P(\tau|p) - \beta \sum_{\tau \in q^{\mathrm{pen}}} \log P(\tau|p) + \gamma\, u_{\mathrm{epi}}(p, s_t)$ over required, preferred, and penalty tags. The four task families are defined by semantic place patterns: water search (target $\{$water$\}$), safe rest ($\{$rest$\}$ with state-conditioning), goal-region rejoin ($\{$goal-region$\}$), hazard recovery ($\{$post-hazard-exit$\}$).

D.3 Single-agent supporting tables and figures

Table 8: Best semantic result per task family from the final 10-seed benchmark, with 95% bootstrap CIs. Hazard-recovery is bottlenecked by retrieval; goal-region remains downstream-bottlenecked.

Task	Best method	Assist	Success (95% CI)	Steps	Query (op.)	Retrieval (op.)	Mat. (op.)	Post (op.)
Water	`water-cp`	on	0.95 [0.88, 1.00]	17.99	0.71	0.71	0.71	0.70
Rest	`rest-s2`	on	0.93 [0.85, 0.99]	23.35	0.61	0.61	0.61	0.73
Goal region	`goal-n1`	off	0.54 [0.52, 0.56]	49.34	0.00	0.00	0.00	0.00
Hazard recovery	`hazard-s7`	on	0.40 [0.35, 0.45]	10.18	0.11	0.11	0.11	0.16

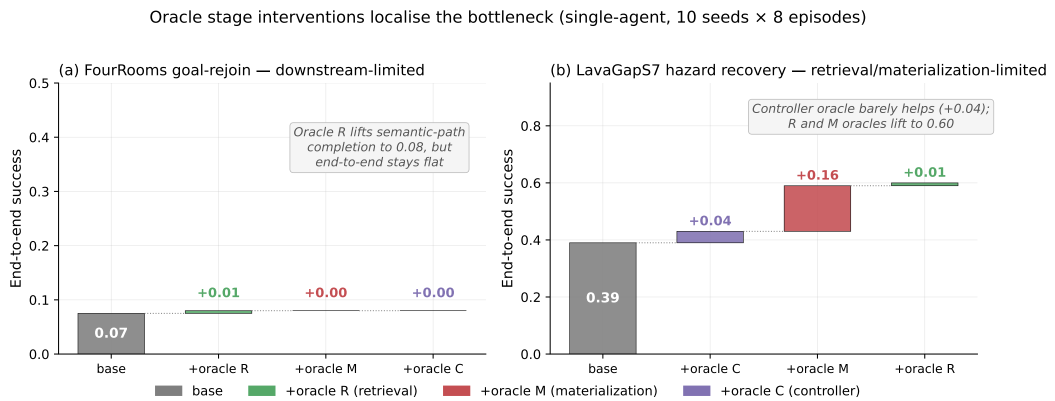

Figure 6: Oracle stage interventions on the two hardest single-agent task families (10 seeds $\times$ 8 episodes). **(a)** FourRooms goal-rejoin: replacing any single stage with an oracle leaves end-to-end success at 0.07; the bottleneck is downstream of the stages in the chain. **(b)** LavaGapS7 hazard recovery: oracle controller adds only $+0.04$, but oracle materialization adds $+0.16$ and oracle retrieval adds a further $+0.01$ to reach 0.60. The framework localises the bottleneck stage-by-stage.

Seed robustness (30 seeds). The single-agent benchmark of §3 uses 10 seeds. To check the diagnosis is not a small-sample artefact, we reran all four task families at 30 seeds, both assist modes, on an independent SLURM cluster (`-num_seeds 30`; e.g. n=240 hazard-recovery and 480 goal-region agent-episodes per mode). Success (assist off): water 0.90, rest 0.84, goal-region 0.61, hazard recovery 0.18; empty-candidate-set rates 0.03, 0.00, 0.37, 0.65 respectively. The *qualitative* attribution is unchanged: hazard recovery stays retrieval-starved (empty-candidate-set rate ≈ 0.62–0.65, low success) while goal-region is far less retrieval-limited (empty-set ≈ 0.37, success ≈ 0.61) — the same stage *ordering* the framework reports at 10 seeds. Absolute rates depend on the exact benchmark configuration (episode budget, assist and env-change settings) and so differ from the headline $0.39/0.91$ of the 10-seed oracle configuration; what 30 seeds confirms is the ordering the diagnosis rests on, not a particular level.

E Portability and scale-up

E.1 BabyAI portability transfer (10 seeds, full benchmark)

A 10-seed portability benchmark on `BabyAI-GoToObjMaze-v0` (Chevalier-Boisvert et al., 2018) (8 episodes per seed, max-steps 200) measured all four single-agent methods used in the main benchmark (random, greedy, `babyai_semantic_node_v1`, `babyai_semantic_plan_v1`) with the same Q/R/M/C logger applied to BabyAI traces. The numbers in Table 9 confirm that (a) the Q/R/M/C events emit non-trivially on the BabyAI substrate, (b) the two semantic variants expose stage-separable structure where the non-semantic baselines have $Q^\star$=$R^\star$=$M^\star$=0 (no semantic queries are issued), and (c) the stage-rate ordering matches the MiniGrid benchmark (Table 2) within bootstrap CIs. GoToObjMaze is a harder task substrate than the LavaGapS7/FourRooms benchmark used in the main text — end-to-end success caps at 0.15 across all methods — which makes the stage decomposition especially informative: the semantic stack is not winning on success but it is the only method family producing measurable per-stage events.

Takeaway. The framework's measurement protocol — not its task-specific results — transfers to a second substrate. End-to-end success on BabyAI is dominated by the much lower episode-level success ceiling of GoToObjMaze, not by the stage structure; conversely, the stage structure that the semantic stack exposes is identical in shape to what the same stack exposes on MiniGrid. This is exactly the substrate-independence property the methodology is intended to provide. Raw data: `tmp/babyai_deeper_10seed/`.

Table 9: BabyAI portability benchmark: 10 seeds $\times$ 8 episodes on `BabyAI-GoToObjMaze-v0`, max-steps 200. The semantic-stack variants succeed at the same end-to-end rate as the greedy baseline (0.15) but are the only methods that produce non-zero Q/R/M/C events — the framework's protocol transfers to the BabyAI substrate without modification. Q/R/M/C operational rates here are evaluated on the same denominator (number of decision points), so $Q{=}R{=}M$ for both semantic variants because GoToObjMaze emits one semantic target type per episode.

Method	Success	Steps	Q (op.)	R (op.)	M (op.)	Post (op.)
random	0.125 ± 0.06	182.7	0.000	0.000	0.000	0.000
greedy	0.150 ± 0.05	171.3	0.000	0.000	0.000	0.000
semantic node	0.150 ± 0.05	171.3	0.032	0.032	0.032	0.025
semantic plan	0.150 ± 0.05	171.3	0.032	0.032	0.032	0.025

E.2 Multi-agent scale-up at $N = 12$

To check that the multi-agent results of §4.5 are not an artefact of the $N \leq 8$ range, an additional 8,640-run sweep was run at $N{=}12$, $T \in \{6, 8, 10\}$, the same three layouts, three hazard densities, eight peer cadences, and 40 seeds. Wall-clock $\approx$ 12 minutes on 8 cores. The conditional-rate signature of §4.5 persists and intensifies (Table 10).

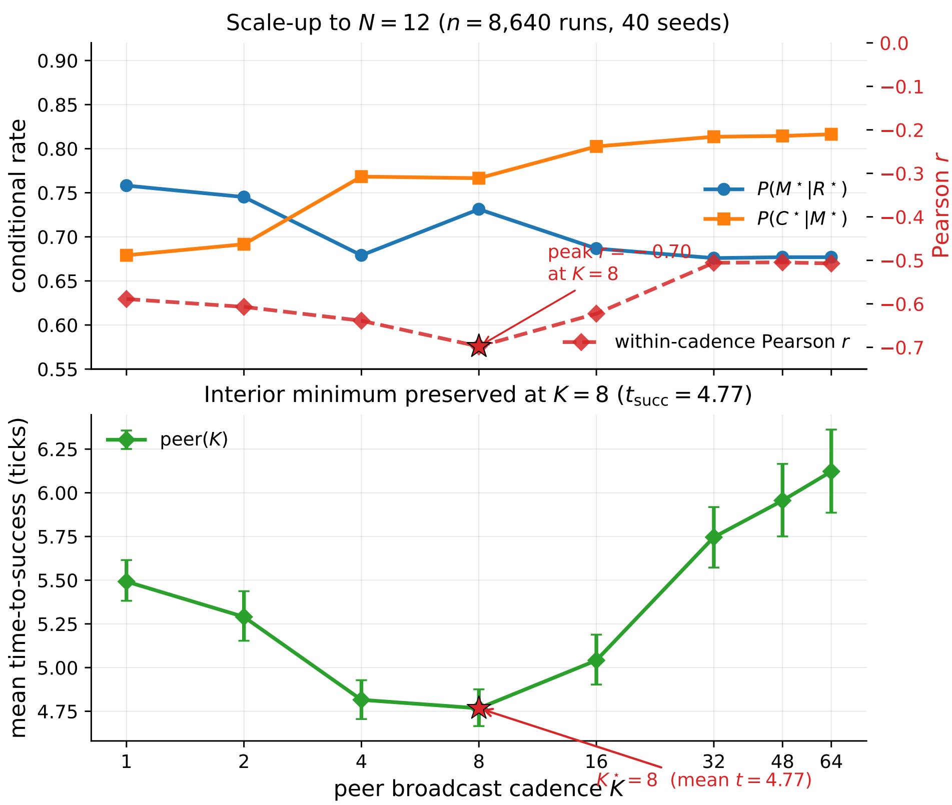

Figure 7: $N{=}12$ scale-up bottleneck signature ($n{=}8{,}640$ runs, 40 seeds). Top: the per-cadence mean rates move in the predicted directions in K; the within-cadence Pearson correlation (red dashed, right axis) peaks at -0.70 at $K{=}8$ (this within-cadence peak is what sharpens at $N{=}12$; the M-side K-slope itself is not cluster-robust at this agent count, Table 10). Bottom: mean time-to-success retains an interior minimum at $K{=}8$ (4.77 ticks); as on the main sweep, the location is robust but the margin is shallow — the gap to the adjacent $K{=}4$ (4.82 ticks) is ≈ 0.05 tick and is not resolved at this sample size. The bottleneck-shift peak migrates outward from $K{=}4$ at $N{\leq}8$ to $K{=}8$ at $N{=}12$, consistent with stronger occupancy coupling at higher agent count delaying the cost-regime transition.

Table 10: $N=12$ scale-up sweep numerical detail (peer architecture; 27 configuration cells). Effect sizes with cluster-robust 95% CIs (block bootstrap over cells): Spearman of $P(C^\star|M^\star)$ with K is $+0.31$ (CI $[+0.21, +0.41]$, excludes zero), while Spearman of $P(M^\star|R^\star)$ with K is only -0.14 (CI $[-0.28, +0.02]$, *not* resolved from zero at N=12). The C-slope of Empirical Claim 1 holds; the M-slope is directionally consistent but not cluster-robust at this agent count. What sharpens at N=12 is the within-cadence correlation and the t_{succ} minimum (Figure 7), not the M-side K-slope.

K	$P(M^\star\|R^\star)$	$P(C^\star\|M^\star)$	MOP	mean t_{succ}	corr
1	0.758	0.679	0.475	5.49	-0.59
2	0.745	0.692	0.474	5.29	-0.61
4	0.679	0.768	0.482	4.82	-0.64
8	0.731	0.767	0.539	**4.77**	$\mathbf{-0.70}$
16	0.687	0.803	0.538	5.04	-0.62
32	0.676	0.814	0.538	5.75	-0.51
48	0.677	0.814	0.539	5.96	-0.51
64	0.677	0.816	0.541	6.12	-0.51

E.3 Q/R/M/C under semantic, frame-free place matching

The main sweep resolves place identity by shared (x, y) keys. To check that the decomposition is not tied to a shared coordinate substrate, we rerun a multi-agent navigation loop in which every agent holds a *private* coordinate frame (a secret per-agent translation) and places are matched *by meaning*: local semantic fingerprint constellations recover the inter-agent frame and transport a peer's target evidence into the receiver's frame (`experiments/warp/semantic_peer_memory.py`; runner `experiments/warp/exp_semantic_cadence_qrmc.py`). Agents navigate with the same rule-based planner toward the transported target; Q/R/M/C are logged unchanged, with $M^\star$ the lock of a candidate within ϵ of a true target (N=6, T=2 scarce water sources, 8 seeds).

The stage events remain well-defined without a shared frame, and the framework localises the coordinate assumption's failure. The *semantic* arm locks a *true* target on its first materialisation in 100% of agent-episodes (at every cadence); the *coordinate* arm — which reads a peer's private coordinates at face value — first-locks a true target only 25–52% of the time (rising with K as agents accumulate their own observations), i.e. it *fails open*, materialising phantom cells that are not water. End-to-end success is scarcity-limited and comparable across arms ($\approx 0.29 \approx T/N$), as expected. This is the diagnostic property of interest: $M^\star$ under coordinate identity fires on non-targets, and the stage logger exposes it.

We do *not* claim the M$\leftrightarrow$C cadence shift here: with an all-to-all broadcast and a 140-tick horizon the semantic frame recovers at every tested K, so $P(M^\star|R^\star)$ is cadence-invariant in this setup (1.00 throughout). The experiment isolates the *place-identity substrate* question — Q/R/M/C is architecture- and coordinate-independent, and detects fail-open materialisation — rather than the cadence result, which stands on the main coordinate-keyed sweep. Folding semantic matching into the scarcity regime that produces the shift is the natural next integration.

F Extended diagnostics

F.1 Conditional-product detail and the Jensen gap

The conditional product $\Phi = P(M^\star|R^\star)\cdot P(C^\star|M^\star)$ in the 35,640-run sweep is monotonically increasing in K over $K \in \{4, 8, 16, 32, 48, 64\}$ and converges to the independent-agent boundary $\Phi(K \to \infty) = 0.605$. Paired bootstrap: $\Delta(K{=}32{-}K{=}16) = +0.008$ (CI $[+0.006, +0.010]$), $\Delta(K{=}48{-}K{=}16) = +0.011$, $\Delta(K{=}64{-}K{=}16) = +0.012$; the independent-agent level sits above every peer K except $K=64$ (where the gap's CI includes zero). This is a statement about the *level* of Φ at the tested cadences, not about the *shape* of $\Phi(K)$: consistent with the monotone trend, we make no claim of an interior extremum (§4.3).

The negative within-cadence covariance documented in §4.5 creates the Jensen-inequality gap between Φ and the marginal product-of-means: $E[X]\cdot E[Y] > E[X\cdot Y]$ when $\mathrm{Cov}(X, Y) < 0$. The mean-of-products is reported as the primary summary because it absorbs the covariance correctly.

Table 11: Per-architecture conditional Q/R/M/C rates from the 35,640-run sweep (40 seeds). $P(R^\star|Q^\star)$ is saturated. POM = product-of-means; MOP = mean-of-products (the proper per-configuration expected product). The two summaries disagree systematically at small/intermediate K because the within-cadence correlation between $P(M^\star|R^\star)$ and $P(C^\star|M^\star)$ is strongly negative there (last column) — the bottleneck-shift signature. **Bold**: MOP-best of the peer cadences. Mean t_{succ} (last column) is the practically relevant efficiency measure; its interior minimum at $K = 8$ is the predicted-shape result, robust in location but shallow in margin (adjacent fast cadences not resolved; Appendix F.3).

Architecture	$P(R^\star\|Q^\star)$	$P(M^\star\|R^\star)$	$P(C^\star\|M^\star)$	POM	MOP (95% CI)	corr	mean t_{succ}
independent	0.99	0.68	0.88	0.598	0.605 [0.597, 0.614]	$+0.00$	8.02
shared bus	1.00	1.00	0.60	0.600	0.595 [0.586, 0.604]	-0.10	8.33
central aggregator	1.00	0.97	0.61	0.592	0.572 [0.564, 0.580]	-0.48	7.64
peer ($K = 1$)	1.00	0.97	0.60	0.585	0.573 [0.564, 0.581]	-0.43	7.80
peer ($K = 2$)	1.00	0.97	0.62	0.594	0.576 [0.568, 0.584]	-0.54	7.59
peer ($K = 4$)	1.00	0.92	0.66	0.599	0.569 [0.561, 0.577]	-0.61	8.00
peer ($K = 8$)	0.99	0.69	0.87	0.597	0.586 [0.578, 0.595]	-0.19	**7.47**
peer ($K = 16$)	0.99	0.67	0.89	0.592	0.590 [0.581, 0.598]	-0.05	7.87
peer ($K = 32$)	0.99	0.67	0.89	0.595	0.598 [0.589, 0.606]	-0.04	8.28
peer ($K = 48$)	0.99	0.68	0.88	0.598	0.602 [0.593, 0.611]	-0.02	8.79
peer ($K = 64$)	0.99	0.68	0.88	0.599	**0.603** [0.594, 0.611]	-0.02	9.19

Figure 8 shows the K-sweep view of both conditional rates and mean t_{succ}; Figure 9 shows the per-configuration scatter that the negative covariance summarises.

F.2 Design decomposition and run-count reconciliation

The 35,640 figure decomposes exactly as in Table 12. The base design is the nine admissible (N, T) scarcity pairs (for N=3: $T\in\{2, 3\}$; N=5: $T\in\{2, 3, 5\}$; N=8: $T\in\{2, 3, 5, 8\}$) crossed with three layouts, three hazard densities, and 40 seeds, giving 3,240 base configurations. The peer architecture is run at all eight cadences (3,240 $\times$ 8 = 25,920); the three cadence-free architectures (independent, shared bus, central aggregator) contribute 3,240 each. The 25,920-run figure used for the peer cluster-robust analysis (Appendix F.3) is exactly the peer block; the initial 12,960-run exploratory sweep (§4.3, "Chronology") and the extended-cadence and N=12 sweeps (Appendix E) are separate run sets and are not pooled with the main sweep.

Table 12: Run-count reconciliation for the main multi-agent sweep (35,640 runs). Counts are recomputed directly from the released run log.

Factor	Levels	Count
(N, T) scarcity pairs	$(3, 2)(3, 3)(5, 2)(5, 3)(5, 5)(8, 2)(8, 3)(8, 5)(8, 8)$	9
Layouts	symmetric, asymmetric, random	3
Hazard densities	$0.0,\ 0.05,\ 0.10$	3
Seeds		40
Base configurations	$9 \times 3 \times 3 \times 40$	3,240
Peer (all 8 cadences K)	3,240 $\times$ 8	25,920
Independent / shared / central	3 $\times$ 3,240	9,720
Total		**35,640**

F.3 Effect sizes and cluster-robust inference

The $p < 10^{-4}$ figures quoted in §4.5 are reported for completeness but are not the basis of the claim: at tens of thousands of runs almost any non-zero slope is "significant," and the runs are not independent—they cluster within the $(N, T, \text{layout}, \text{hazard})$ configuration. The full sweep has 25,920 peer runs but only **81 independent design cells**. We therefore report effect sizes with cluster-robust confidence intervals from a block bootstrap that resamples *configuration cells* (not rows), B=4000 resamples.

The two slopes of Empirical Claim 1 survive as genuine, but moderate, effects: the Spearman correlation of $P(M^\star|R^\star)$ with K is $\rho = -0.53$ (95% cluster CI $[-0.59, -0.47]$, strong) and of

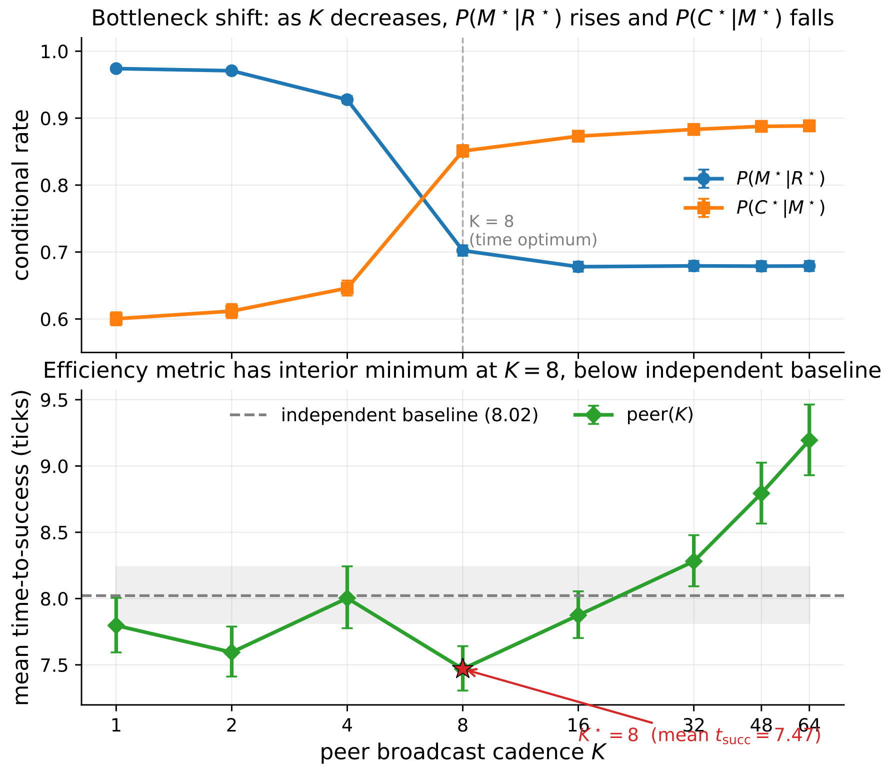

Figure 8: Bottleneck shift across eight peer cadences (40-seed sweep). Top: $P(M^\star|R^\star)$ falls and $P(C^\star|M^\star)$ rises in K (moderate, cluster-robust Spearman effect sizes -0.53 and $+0.43$; CIs in Appendix F.3). Bottom: mean time-to-success has an interior minimum at $K = 8$ (7.47 ticks) below the independent-agent baseline (8.02, dashed); the location is robust but its margin over the adjacent fast cadences (K=1, 4) is not resolved (Appendix F.3). Error bars and shaded band are 95% bootstrap CIs.

$P(C^\star|M^\star)$ with K is $\rho = +0.43$ (95% cluster CI $[+0.37, +0.50]$, moderate). Both intervals exclude zero under cell-level resampling, so the bottleneck shift is not an artefact of the correlated run count; it is a moderate monotone trend, and we describe it as such rather than as a near-deterministic law.

The interior optimum on mean time-to-success is robust in *location* but shallow in *margin*. Paired block bootstrap on the cell-level t_{succ} differences gives $t_{\mathrm{succ}}(K{=}8)$ below $t_{\mathrm{succ}}(K{=}16)$ by 0.40 ticks (CI $[0.08, 0.83]$) and below $K{=}64$ by 1.67 ticks (CI $[0.75, 2.83]$), both resolved; but its advantage over $K{=}4$ (0.49, CI $[-0.47, 1.51]$) and $K{=}1$ (0.27, CI $[-0.43, 0.97]$) is within noise.[1] A success-weighted efficiency metric (t_{succ} divided by success rate) has the same interior $\arg\min$ at $K{=}8$, so the location of the optimum is robust to the efficiency definition even though its margin over the adjacent fast cadences is not resolved at this sample size. We read this as support for the *shape* predicted by the M↔C trade-off, not for a sharply tuned optimal cadence. Two further cell-paired contrasts complete the picture. On t_{succ}, $K{=}8$ is also resolved against the independent baseline (-0.40, CI $[-0.59, -0.24]$). On a censoring-aware expected time $\mathbb{E}[T] = s\, t_{\mathrm{succ}} + (1 - s)\, T_{\max}$ (failed runs charged the step limit $T_{\max}{=}120$), the fast flank reverses: $K{=}8$ beats $K{=}4$ (-2.10, CI $[-3.61, -0.72]$) but loses to $K{=}16$ ($+0.69$ $[+0.09, +1.43]$) and to independent ($+0.32$ $[+0.05, +0.63]$). The interior optimum is thus a property of conditional speed among successes; on unconditional efficiency the completion collapse of fast sharing dominates, which is the gap the reservation protocol of open problem (3) closes.

[1]This paired bootstrap and the effect-size Spearman correlations above are computed on the same sweep execution as Table 11 (25,920 peer runs); the t_{succ} statistic is defined only over successful episodes, so the paired cell-level difference (1.67 for $K{=}8$ vs. $K{=}64$) is on that subpopulation and matches the marginal-mean difference in Table 11 ($9.19{-}7.47{=}1.72$) to within rounding. The effect sizes are identical on the independent `paper1_clean` execution ($-0.531/{+0.434}$), confirming they are not run-specific.

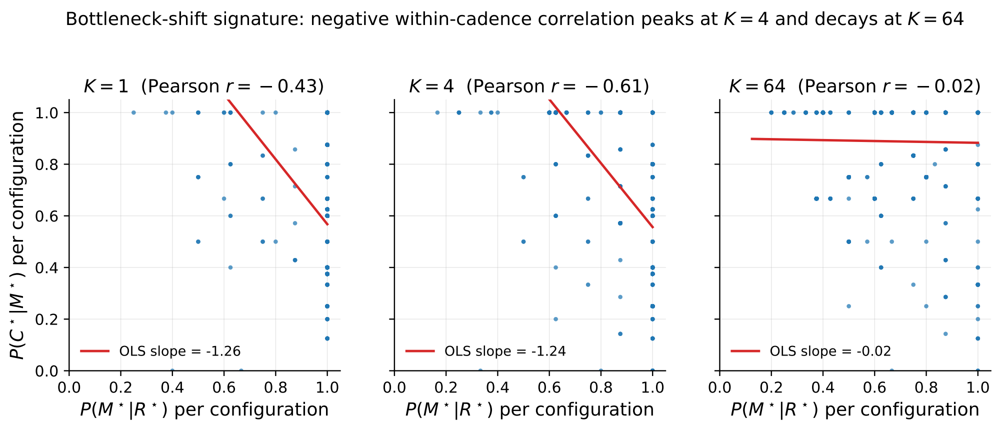

Figure 9: Per-configuration scatter of $P(M^\star|R^\star)$ vs. $P(C^\star|M^\star)$ at three cadences. The strong negative correlation at $K = 4$ (Pearson -0.61) is the bottleneck-shift signature: the same configurations that succeed at M tend to fail at C and vice versa. The correlation decays to noise by $K = 64$.

F.4 Construct validity: task success Y vs. semantic-path completion $C^\star$

A success Y can in principle bypass the semantic path, so a stage bottleneck in $Q/R/M/C$ need not explain variation in Y. Two facts bound this concern. First, in the multi-agent sweep that Empirical Claim 1 is about, success is *defined* as reaching a `water_source` target, which is exactly the $C^\star$ event; empirically Y and $C^\star$ coincide run-for-run (correlation 1.00, mean gap 0.00). The central multi-agent claim is thus not exposed to a parallel-channel objection. Second, the gap $Y > C^\star$ is a single-agent phenomenon on the easy task families (water, rest), where a fallback controller can reach the goal without a semantic target lock; it is largest exactly where success is already high and the diagnosis is least needed, and it vanishes on the hard families (hazard recovery, goal-region) where the framework does its work. A per-episode, stage-resolved mediation of Y (rather than of $C^\star$) requires logs that tag each success with whether it traversed the semantic path, and is a planned instrumentation refinement.

F.5 Threshold sensitivity of the Q/R/M/C event definitions (θ and ϵ)

The single-agent attribution "candidate generation fails, not ranking" (§3.5) rests on a high empty-candidate-set rate. A reviewer asks whether that is an artefact of the required-tag threshold θ (deployed 0.5; the candidacy gate is $\text{profile}[\text{tag}] \geq \theta$ in `symbolic_memory._required_matches_profile`). We sweep θ via an override on the query builder (`experiments/big_experiment/theta_ablation.py`); the endpoints confirm the gate is live, and the operating range shows the diagnosis is not a threshold effect.

Table 13: Empty-candidate-set rate vs. required-tag threshold θ (semantic method, assist off, hazard-recovery / goal-region). The rate is *invariant* across the entire realistic range $\theta \in [0.05, 0.95]$; only the degenerate endpoints move it, and only to the no-candidate floor. The empties are therefore absence of stored evidence, not sub-threshold filtering.

θ	empty rate (hazard recovery)	empty rate (goal region)
0.00 (accept all)	0.50	—
0.05	0.533	0.267
0.15	0.533	0.267
0.25	0.533	0.267
0.50 (deployed)	0.533	0.267
0.75	0.533	0.267
0.95	0.533	0.267
1.00 (accept none)	0.75	—

The reading is decisive (Table 13). At θ=1.0 nothing passes and the empty rate rises; at θ=0.0 the required-tag gate accepts every node ($\mathrm{conf} \geq 0$ always holds), yet the empty rate falls only to 0.50, its floor—meaning that at those decision points there is *no candidate node of any confidence* to retrieve. Across the entire sane range $\theta \in [0.05, 0.95]$ the rate is flat at 0.533: loosening or tightening the threshold does not recover a candidate, because the required semantic place was never stored. This is exactly the "absence of evidence, not contradiction" regime named in §2.3, now shown to be threshold-invariant, so the memory-accumulation attribution is not an artefact of θ. (Absolute rates here are on a small hazard-recovery / goal-region sweep and are not the headline 91% of §3.5, which is on the full benchmark; the point is the θ-invariance, not the level.)

The match/lock tolerance ϵ. The other free constant is ϵ, the tolerance with which a materialised lock counts as "within a true target" (the $R^\star$/$M^\star$ match and lock definition, §2.3). Sweeping $\epsilon \in \{0.3, 0.6, 1.0, 1.5\}$ on the scarcity cell (N=8, T=3, 15 seeds, peer, `experiments/big_experiment/exp_eps_sensitivity.py`) leaves the two slope signs of Empirical Claim 1 intact: $\mathrm{Spearman}(P(M^\star|R^\star), K)$ is $-1.00, -1.00, -0.87, -0.63$ and $\mathrm{Spearman}(P(C^\star|M^\star), K)$ is $+0.50, +0.50, +0.87, +0.87$ across the four tolerances. A looser ϵ raises absolute $P(M^\star|R^\star)$ (more locks count as on-target) but does not flip either slope. So the bottleneck shift is robust to both free thresholds of the logger: θ (candidacy/gating, above) and ϵ (match/lock).

F.6 Robustness to Assumption 1 violations

Definition 1 rests on the conditional stage-Markov property (Assumption 1). The two practically relevant violations—adaptive query rewrites not surfaced to the logger, and hidden between-stage memory updates—are exactly cases where an *unlogged latent* couples non-adjacent stages. We quantify the resulting estimator bias with a controlled simulation (4×10^5 episodes per setting; `experiments/big_experiment/assumption1_stress_test.py`). Episodes are generated with known mechanism rates $P(M^\star|R^\star)$=0.70, $P(C^\star|M^\star)$=0.65, and a hidden binary latent U that shifts *both* the M-rate and the C-rate by $\pm\delta$; the naive estimator conditions only on the previous starred stage, not on U.

Table 14: Bias of the naive conditional estimators under an unlogged latent of strength δ that couples the M and C stages (violating Assumption 1). $P(M^\star|R^\star)$ stays unbiased; $P(C^\star|M^\star)$ is biased upward, materially only under strong coupling.

δ (coupling strength)	bias in $\hat{P}(M^\star|R^\star)$	bias in $\hat{P}(C^\star|M^\star)$
0.00	−0.001	−0.002
0.05	+0.001	+0.004
0.10	+0.001	+0.015
0.15	−0.000	+0.031
0.20	+0.000	+0.056
0.25	−0.001	+0.089

Two conclusions (Table 14). The R→M conditional is unbiased at every δ: the latent re-weights episodes but does not confound that edge. The C→M conditional acquires a one-signed *upward* bias—the latent makes high-M episodes also high-C, so conditioning on M over-counts completion—but the bias is only $+1.5$ percentage points at a mild violation ($\delta \leq 0.10$) and reaches $+5.6$ points only under strong unlogged coupling (δ=0.20). The framework's exposure to Assumption 1 violations is therefore bounded and its failure direction is known, which is the relevant robustness statement for applying the logger to learned or LLM agents where such couplings are more common.

F.7 Continuous-substrate validation (VMAS): both slopes with cluster-robust CIs

The most-requested robustness check is whether the M↔C bottleneck shift is specific to grid substrates. We validate it on a continuous-state substrate (VMAS (Bettini et al., 2022)) at scale: three scarcity cells $(N, T) \in \{(6, 2), (8, 3), (10, 4)\}$ × five cadences $K \in \{1, 2, 4, 8, 16\}$ × 40 seeds (4,800 agent-episodes; `experiments/vmas_portability/run_vmas_full.py`, executed on a SLURM CPU cluster). The symbolic peer memory and the operational Q/R/M/C definitions are reused unchanged; a thin discretisation bridge maps continuous (x, y) to coarse cells so the memory applies, and the controller acts in continuous space toward the materialised waypoint. Materialisation uses the same occupancy-sensitive lock as the grid planner.

Table 15: VMAS continuous-substrate validation of Empirical Claim 1: per-cell conditional rates and block-bootstrap effect sizes (B=2,000 resamples over the 120 (cell, seed) blocks). Both slope signs hold in *every* scarcity cell, and the bootstrap CIs exclude zero decisively: $\mathrm{Spearman}(P(M^\star|R^\star), K) = -0.999$, 95% CI $[-1.000, -0.975]$; $\mathrm{Spearman}(P(C^\star|M^\star), K) = +0.995$, 95% CI $[+0.900, +1.000]$. Rates are operational per-event ratios (the $(10, 4)$, K=1 M-ratio marginally exceeds 1 because a lock can fire without a true-target retrieval event; shown capped).

(N, T)	K=1		K=4		K=16	
	$P(M^\star\|R^\star)$	$P(C^\star\|M^\star)$	$P(M\|R)$	$P(C\|M)$	$P(M\|R)$	$P(C\|M)$
$(6, 2)$	1.00	0.22	0.51	0.41	0.27	0.74
$(8, 3)$	1.00	0.18	0.45	0.41	0.28	0.64
$(10, 4)$	1.00	0.18	0.52	0.38	0.30	0.66

The earlier single-configuration check reported here as preliminary is now superseded: with three scarcity cells, 40 seeds, and the same block-bootstrap methodology as the main sweep (Appendix F.3), both slope signs of Empirical Claim 1 transfer to continuous positions and motion with resolved effect sizes, in every cell. The freshness-vs-contention mechanism is therefore not an artefact of grid cells. Remaining honest limits: the memory is fed through the discretisation bridge (continuous *dynamics* with discretised symbolic *memory*), the horizon is short (25 steps), the robots are holonomic points, and the sweep covers the coupling regime $K \leq 16$; visually rich 3D substrates (Habitat, ProcTHOR) remain future work.

F.8 External language-agent substrate: ALFWorld

To take the contract beyond our own substrates, we instrument standard ALFWorld (Shridhar et al., 2021) text games (ALFRED household tasks, `eval_out_of_distribution` split) with an LLM policy and the same four-stage contract: $Q^\star$ fires when the LLM declares the task's target object; an episodic object→receptacle memory is built from observations; $R^\star$ fires when that memory holds a candidate location for the declared target; $M^\star$ when the agent commits (`go to`) to the retrieved candidate; $C^\star$ on environment success (`experiments/alfworld_qrmc/run_alfworld_qrmc.py`; HF backend, greedy, 25 episodes, step limit 40, executed on a SLURM GPU node).

On Qwen2.5-3B-Instruct the stage taxonomy is sharply informative even at zero end-to-end success: $Q^\star$=1.00 (the model always extracts a target), $R^\star$=0.16, $M^\star$=0.00, $C^\star$=0.00, with first-missing-stage counts 21× R_fail and 4× M_fail over 25 episodes. A success-only evaluation would report "0.00" and stop; the decomposition reports *where* the 3B agent loses the task: predominantly it never grounds the declared object to a location (R), and when it does, it fails to commit to the retrieved candidate (M). This matches the near-zero ALFWorld scores reported for small models in the literature while adding the stage attribution they lack. (Instrumentation honesty: a first 7B run landed on an 11 GB GPU and out-of-memoried on every call; it is excluded and rerun on a 24 GB node — the cached error strings made the failure trivially auditable.) Scaling this external validation to larger models and more episodes is mechanical; the point established here is that the identical event contract instruments an external, community-standard language-agent benchmark and yields non-trivial stage attributions on real failures.

F.9 Cadence-robustness: the shift tracks scarcity, not grid design

A reviewer may object that the bottleneck shift is built into a scarce-target grid with locks. If instead it is a signature of contention over *scarce shared* targets, its slope conditions must break when targets are abundant. We rerun the peer cadence sweep in three regimes on the same env (N=8, random layout, hazard 0.05, 20 seeds; `experiments/big_experiment/exp_cadence_robustness.py`): scarcity (T=3), borderline (T=N=8), and abundance (T=14>N).

Both slopes are present under scarcity, weaken at the border, and *flatten* under abundance: with more targets than agents there is no contention, so fast sharing neither over-saturates materialization ($P(M^\star|R^\star)$ stays flat at ~ 0.75 instead of spiking then falling) nor collapses completion ($P(C^\star|M^\star)$ stays at ~ 1 even at K=1 instead of dropping to 0.51). The within-cadence M↔C anticorrelation

Table 16: The two slope conditions of Empirical Claim 1 (§4.3) hold under scarcity and flatten under abundance. $P(M^\star|R^\star)$ should fall and $P(C^\star|M^\star)$ should rise across K; "corr" is the within-cadence Pearson correlation of the two links at $K{=}64$ (the shift signature). $P(C^\star|M^\star)$ is an operational ratio (capped at 1; abundance completes without a formal lock).

Regime	$P(M^\star\|R^\star)$: $K{=}1{\to}64$	$P(C^\star\|M^\star)$: $K{=}1{\to}64$	corr(M, C) @ $K{=}64$	t_{succ} argmin
Scarcity ($T{=}3$)	0.86 → 0.49 (falls)	0.51 → 0.79 (rises)	-0.86	$K{=}4$ (interior)
Borderline ($T{=}N{=}8$)	0.79 → 0.72	0.77 → 0.95 (rises)	-0.79	$K{=}8$ (interior)
Abundance ($T{=}14$)	0.74 → 0.77 (flat)	1.00 → 0.99 (flat)	-0.29	$K{=}4$

attenuates monotonically with abundance ($-0.86 \to -0.79 \to -0.29$). The shift therefore tracks the *contention mechanism* and vanishes when it is removed — it is not a fixed artefact of the grid-plus-locks design. We note honestly that the interior t_{succ} optimum is more general (present in all three regimes): it reflects the exploration/travel cost of very fast *or* very slow sharing, of which target contention is one contributor but not the only one.

F.10 Open problems suggested by the diagnostics

Four concrete questions follow directly from the measurements of §3.5 and §4.5. **(1) Adaptive cadence.** Empirical Claim 1 assumes a fixed K; the bottleneck-shift signature suggests an online controller K_t that broadcasts when the (estimated) M-side gain exceeds the C-side occupancy cost. **(2) Selective consolidation.** Snapshot exchange merges everything; merge rules that prioritize evidence about under-represented semantic tags attack the candidate-generation starvation of §3.5 directly. **(3) Trust-weighted merging under occupancy.** The central aggregator already performs a trust-weighted merge but pays the C-collapse cost; decentralized trust rules that account for peer *intent* (not only peer evidence) could decouple freshness from target racing. *Since submission of the first draft, this problem has been addressed constructively*: a decentralized protocol in which an agent's target lock piggybacks a lightweight *reservation* on its next scheduled broadcast (no coordinator, no extra channel) and locks dominated by an earlier reservation are retracted by an anti-$M^\star$ rollback with exponential backoff. On the scarcity cells this recovers the fast-cadence completion loss on *unconditional* metrics — success rate rises at every $K \in \{1, 2, 4\}$ with disjoint bootstrap CIs, and fast sharing with the protocol exceeds the $K{=}8$ optimum baseline (0.614–0.620 vs. 0.583) — with rollback latency bounded by $\approx K$ ticks. Details in a companion manuscript; the open problem stands revised toward its harder variants (adaptive reservation scope, heterogeneous trust). **(4) Substrate generality.** The decomposition transfers across MiniGrid, BabyAI, the custom grid, and a standard-MiniGrid multi-agent wrapper with identical operational definitions and verified slope signs of Empirical Claim 1 (§ 6), and a preliminary check reproduces both slopes on the continuous-state VMAS substrate (Appendix F.7). The decomposition is also coordinate-independent: under *semantic*, frame-free place matching (agents in private frames, places matched by meaning) the stage events remain well-defined and expose the coordinate assumption's fail-open materialisation (Appendix E.3). A full continuous-state validation with cluster-robust CIs, and richer 3D substrates (Habitat (Savva et al., 2019)), remain the next portability steps.

F.11 Extended related work

Classical cognitive-map perspectives motivate explicit place abstractions (Tolman, 1948; O'Keefe and Nadel, 1978; Kuipers, 2000); modern embodied AI combines mapping and planning (Gupta et al., 2017) or exposes topological memory (SPTM, Savinov et al., 2018); goal-conditioned RL conditions on explicit goals or embeddings (Andrychowicz et al., 2017; Ghosh et al., 2021), with successor-representation work providing predictive structure (Dayan, 1993; Stachenfeld et al., 2017; Momennejad et al., 2017). Our setting differs in that the target is a semantic pattern retrieved from memory rather than a coordinate. In multi-agent RL, CTDE dominates (Lowe et al., 2017; Foerster et al., 2018) and learned communication adds a per-step exchange variable (Sukhbaatar et al., 2016; Foerster et al., 2016); gossip and consensus protocols (Boyd et al., 2006; Xiao et al., 2007) are the closest network-level analogue of our cadence axis, lifted here to the cognitive-architecture level. LLM-agent systems combine reasoning/acting, reflection, explicit memory, and multi-agent conversation (Yao et al., 2023; Shinn et al., 2023; Wang et al., 2023; Park et al., 2023; Wu et al., 2023; Packer et al., 2023; Chhikara et al., 2025). Our framework is orthogonal: rather than learning a communication policy end-to-end, it *diagnoses* where in the Q/R/M/C chain a given architecture succeeds or fails, and yields a slope-level prediction about cadence that is testable on logs.

Stage-wise evaluation. The closest line decomposes retrieval-augmented generation: RAGAS scores retrieval and generation quality separately (Es et al., 2024), RAGChecker refines this to claim-level diagnostics meta-evaluated against human judgments (Ru et al., 2024), and AgentBoard replaces final success with analytic progress rates for multi-turn LLM agents (Ma et al., 2024). Q/R/M/C differs on three axes: its events are logged actions of a task-executing agent, not LLM-judged text quality; it models commitment (M) and the multi-agent coupling channels (shared memory, occupancy) that RAG metrics do not represent; and its diagnoses are validated interventionally (oracle stages; engineered structural faults) rather than by correlation with human ratings. Process mining and conformance checking compare event logs against stage models (van der Aalst, 2016; Carmona et al., 2018); the protocol shares that log-level discipline and adds agent-specific event semantics with intervention-based validation. Process reward models score intermediate reasoning steps to supervise training (Lightman et al., 2024). The stance of specifying and checking externally observable events rather than internal state is standard in runtime verification and distributed-system observability (Leucker and Schallhart, 2009); Q/R/M/C applies it to the memory interface of task-executing agents. Closest in mechanism is intervention-supported error attribution: REFLECT localizes a candidate error step in an LLM agent trace and verifies it through controlled replay with a diagnosis-specific patch (Lin et al., 2026). Q/R/M/C differs in unit and in cost: it measures recurring pipeline stages and their conditional rates from logs alone, and uses interventions (oracle stages; stage-level repairs) to validate the protocol rather than to produce each individual diagnosis.

F.12 SOTA baseline and coverage matrix

Table 18 separates direct experimental baselines from broader SOTA references. The local experiments are intentionally modest and fully instrumented: random / graph-only / SPTM-like / learned BC in MiniGrid, CommNet PPO in the multi-agent grid, and a TextNav LLM step-budget study over two local Ollama models (in which Q/R/M/C localizes a completion-limited failure, §4.5). ReAct, Reflexion, Voyager, Generative Agents, AutoGen, MemGPT, and Mem0 are not reimplemented as full systems here; they define the current memory-agent design space against which Q/R/M/C is positioned as a diagnostic protocol.

Positioning against SOTA memory designs. The memory-agent design space can be organised on two axes: single- vs. multi-agent, and how memory is represented and shared (Table 17). Vector-embedding memory (MemGPT, Voyager single-agent; Generative Agents multi-agent) retrieves by opaque nearest-neighbour lookup, so a failure is not localisable to a stage. Shared-context designs (MetaGPT, AutoGen) pool everything into one window, dissolving per-agent state. Learned-communication MARL (CommNet, and stronger CTDE methods QMIX (Rashid et al., 2018), MAPPO (Yu et al., 2022), MADDPG (Lowe et al., 2017)) optimises a message channel end-to-end and, as we verify for CommNet PPO (§4.5), emits no separable query/lock/completion events. The cell our system occupies—per-agent *symbolic* memory with an explicit query/retrieve/materialise interface, exchanged at a tunable cadence—is empty in this matrix, and it is exactly the interface that makes the four Q/R/M/C stages observable. Our claim against SOTA is therefore not higher reward but higher *diagnosability at matched task success*: the same interface that a strong learned method trades away for end-to-end optimisation is what lets the framework localise where memory helps.

F.13 Multi-agent oracle-intervention numerics

F.14 Oracle C: definition and reading

Oracle C is the completion-stage analogue of oracle R and oracle M, added to close the R/M/C triple symmetrically. The real Q/R/M pipeline runs unchanged (real memory query, real materialisation), and once an agent has locked a target that is a genuine `water_source`, the oracle guarantees arrival by placing the agent on that cell; scarcity is preserved because a cell already claimed or occupied is not available, so oracle C cannot make more than T agents succeed. The runner is `experiments/big_experiment/exp_oracle_interventions.py` (mode `oracle_C`).

Two readings follow from Table 19. (i) Oracle C produces the *lowest* mean t_{succ} of the three interventions ($\sim$ 6.1–6.4 vs. $\sim$ 7.2 for R and M), because it removes the entire post-lock travel

Table 17: Where the framework sits in the memory-agent design space. Q/R/M/C stage-observability requires an explicit query/lock interface, which the symbolic-per-agent cell provides.

Memory representation	Single-agent	Multi-agent
Vector-embedding memory	MemGPT, Voyager (Packer et al., 2023; Wang et al., 2023)	Generative Agents (Park et al., 2023)
Shared context window	—	AutoGen (Wu et al., 2023)
Learned-comm. MARL	—	CommNet, QMIX, MAPPO, MADDPG (Sukhbaatar et al., 2016; Rashid et al., 2018; Yu et al., 2022; Lowe et al., 2017)
Symbolic per-agent + explicit interface	Q/R/M/C single-agent (§3)	**this work** (peer memory, cadence K)

time; this confirms that a real part of the excess time is controller travel to a correctly materialised target. (ii) Oracle C yields the *smallest* success-rate gain ($+0.01$–0.02 over baseline, below both oracle R at 0.614 and oracle M at 0.602), because in scarcity the constraint on *whether* an agent succeeds is upstream—retrieving and locking a still-available target—not reaching it. The two facts together are exactly the framework's thesis in intervention form: completion is where time is spent, retrieval/materialisation is where success is won.

F.15 Stage-agnostic cadence hypotheses and what the framework changes

A first version of the multi-agent cadence sweep (12,960 runs at 20 seeds; subsumed by the 35,640-run sweep in this paper) was originally planned with four operational hypotheses formulated *before* Empirical Claim 1. They are retained here because they motivated the eventual framework. The framework was formulated after the sweep was complete; it is not claimed to have predicted the sweep numbers a priori (see "Chronology" in §4.3).

Original hypotheses (pre-framework).

H1. An interior $K^\star \in \{2, 4, 8\}$ minimising mean time-to-success exists for most (N, T, layout) cells.

H2. $K^\star$ grows monotonically with scarcity $\rho = N/T$.

H3. When $\rho \le 1$, $K^\star = 1$.

H4. Peer-broadcast architectures Pareto-dominate centralized aggregation on (mean time-to-success, success rate).

Stand-alone result on the original 20-seed sweep. The practical claim was directionally supported: peer at the empirically best K was faster than centralized in the scarcity regime (a modest effect; Mann–Whitney $p{=}0.044$ on this small standalone sweep, reported as exploratory rather than as primary evidence, consistent with the effect-size-first line of Appendix F.3). Among the four structural hypotheses, only H4 was clearly supported in the random-layout subset. H1 held in 7 of 18 cells; H2 had a non-significant negative correlation; H3 was supported in 4 of 9 admissible cells. The 35,640-run extended sweep used elsewhere in the paper produces matching stand-alone numbers within bootstrap uncertainty.

What the framework changes. The hypotheses H1–H4 are stage-agnostic: each ties $K^\star$ to scarcity ρ without specifying mechanism. Empirical Claim 1, formulated after the sweep, ties the slopes of $P(M^\star|R^\star)$ and $P(C^\star|M^\star)$ to cadence; the slope conditions are testable on logs. The data verify both signs as moderate, cluster-robust effect sizes (Appendix F.3). We make no formal claim about the shape of the conditional product Φ, on which the data are monotone. The interior optimum holds on the efficiency metric mean time-to-success.

G Assist on/off analysis

Across all four single-agent task families, top-line success is essentially unchanged under assist on/off ($\Delta \le 0.10$). The only non-trivial effect is on rest, where assists improve post-retrieval completion by $+0.05$ without changing query/retrieval rates, consistent with assists acting on materialization and post-retrieval execution.

H Retrained retriever and CommNet baseline

H.1 Retrained candidate-generation MLP (§3.5)

Dataset extraction. From the 10-seed hazard-recovery benchmark (MiniGrid-LavaGapS7-v0, method `milestone_state_conditioned_hazard_recovery_v7`), an 18-dimensional feature vector was extracted for every step of every episode of every seed — 3 numeric agent-state scalars (energy, thirst, risk_budget), 9 semantic-tag confidences, 6 intent one-hot — and a binary label: *was this symbolic node ever selected as a candidate during the run?* Dataset totals: 814 examples, 113 positives (13.9%).

Split. By seed: train $\{2, 3, 4, 5, 6, 7\}$ (505 examples, 9.7% positive), val $\{8, 9\}$ (170, 22.4% positive), test $\{10, 11\}$ (139, 18.7% positive).

Architecture. $18 \to 32 \to 32 \to 1$, ReLU activations, BCE with pos_weight ≈ 9.3 for class imbalance, Adam lr $= 2{\cdot}10^{-3}$ with weight_decay $= 10^{-4}$, batch 64, 200 epochs, best by val loss.

Results.

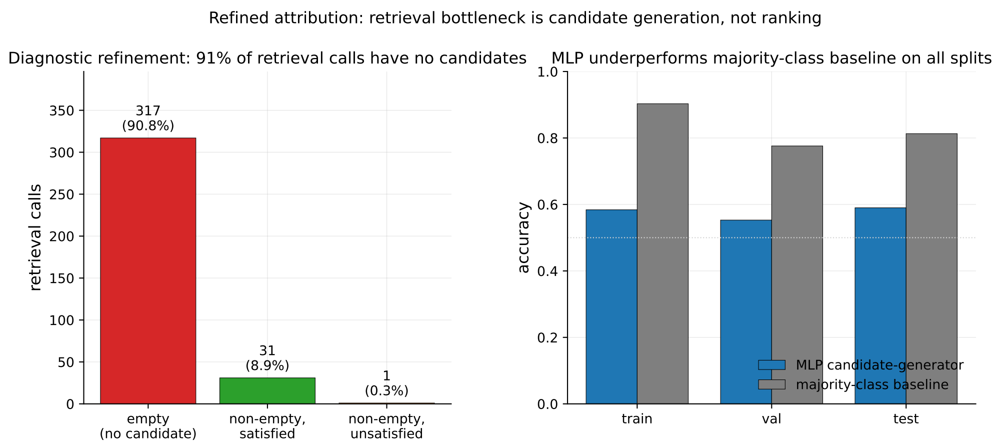

Figure 10: Refined attribution diagnostic. *Left:* 349 retrieval calls aggregated across 10 seeds; 317 (90.8%) returned empty candidate sets, 31 (8.9%) returned at least one satisfied candidate, only 1 (0.3%) returned candidates that were ranked but unsatisfied. The bottleneck is candidate generation, not candidate ranking. *Right:* the trained MLP underperforms a majority-class baseline on all three seed-disjoint splits, confirming that instantaneous state features alone are insufficient to predict candidate eligibility.

The MLP underperforms majority-class on accuracy across all splits and achieves only moderate precision at fixed recall. Instantaneous state features are insufficient to predict candidate eligibility downstream. This is the negative-but-informative finding referenced in §3.5: the framework's "retrieval-limited" attribution refines to "memory-accumulation-limited" rather than "ranking-limited".

H.2 CommNet-style learned communication baseline (§4.5)

Policy architecture. Per-agent observation is 4-dim direction one-hot + 5×5×5 local cell-type window (125) + 2-dim normalised position = 131. Shared-weights encoder 131→64→64 (ReLU), 16-dim comm projection, peer mean-pool (excluding self), combine 80→64 (ReLU), actor head $64 \to 4$ (TURN_LEFT, TURN_RIGHT, FORWARD, NOOP), critic head $64 \to 1$. This is a CommNet-style learned-communication architecture (Sukhbaatar et al., 2016).

Training. REINFORCE (Williams, 1992) with value-baseline; loss $= -\,\mathrm{mean}(\log \pi(a) \cdot (R - V)) + 0.5 \cdot \mathrm{MSE}(V, R) \,-\, 0.01 \cdot H[\pi]$; Adam lr $= 3{\cdot}10^{-4}$, gradient clipping 1.0. Reward shaping: $+1.0$

on first success per agent, -0.01 step cost, -0.1 per hazard contact. 2000 episodes cycling 200 environment seeds, wall-clock 100 s on CPU. Held-out eval: 50 environment seeds 250–299.

Training curve. Episode-level success (last-100 window) stays in $[0.13, 0.23]$ throughout training; no monotonic improvement after $\sim$200 episodes. The policy converges to a local optimum where one of three agents finds water by chance.

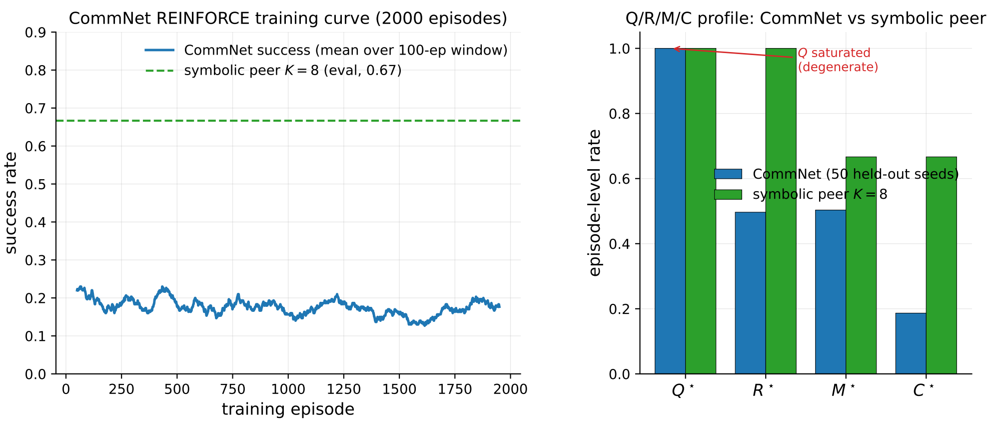

Figure 11: Learned-communication baseline diagnostics. *Left:* CommNet training curve (smoothed success rate over a 100-episode window) over 2000 training episodes; success stays below 0.25 throughout. The dashed line is the symbolic peer architecture at $K = 8$ on the same scenario for reference. *Right:* Q/R/M/C profile of the trained CommNet on 50 held-out seeds vs symbolic peer at $K = 8$. $Q^\star$ is saturated at 1.0 on CommNet (degenerate by construction); R/M collapse together; only on the symbolic architecture are the conditional rates separated.

Q/R/M/C evaluation (50 held-out seeds, stochastic action sampling).

Compare to the symbolic peer architecture at K=8: $P(M^\star|R^\star)$=0.69, $P(C^\star|M^\star)$=0.87, success ~ 0.65.

PPO variant: structural degeneracy verified at competitive success. A PPO-trained (Schulman et al., 2017) variant of the same CommNet architecture (clip ϵ=0.2, generalized advantage estimation (Schulman et al., 2016) with λ=0.95, γ=0.99, 4 PPO epochs per rollout, 64 rollouts per update, 200 updates, $\approx$ 96 minutes on a single CPU core; full setup in `experiments/commnet_ppo_baseline/train_ppo_commnet.py`) reaches 0.667 success on 100 held-out seeds — competitive with the symbolic peer architecture at K=8 (0.65).

The PPO result converts the structural argument from an assertion ("training budget would not change this") to a measurement: $Q^\star$ remains saturated at 1.00, $P(M^\star|R^\star)$ remains 1.00 (R/M collapse), even at competitive success. The Q stage is degenerate by construction in any policy whose communication channel is a real-valued vector with a learned linear projection — the channel is never informatively empty, regardless of training budget — and without an explicit symbolic target-lock interface the M event cannot separate from R. Only architectures with an explicit *query/lock* interface (the symbolic stack of §4.1) expose stage-separable Q/R/M/C events.

MAPPO: the collapse persists on a stronger CTDE baseline. To rule out that the observability collapse is an artifact of CommNet's capacity, we train MAPPO (Yu et al., 2022) — PPO with decentralized actors and a *centralized critic* $V(s)$ over the concatenation of all agents' observations (CTDE) — on the same scarcity scenario (`experiments/mappo_baseline/train_mappo.py`; 300 updates $\times$ 64 rollouts, $\approx$ 8.5 h on a cluster CPU node) and evaluate with the identical Q/R/M/C logger on 50 held-out seeds. MAPPO reaches 0.667 success — competitive with the symbolic peer architecture at this cell — yet its stage profile is structurally degenerate in exactly the CommNet way: $Q^\star$ saturated at 1.00, $P(R^\star|Q^\star)$=0.98, $P(M^\star|R^\star)$=1.00 (R and M collapse into one event), $P(C^\star|M^\star)$=0.69. The centralized critic changes the training signal, not the interface: without an explicit query/lock boundary there is nothing for the M event to separate from R. This upgrades the claim from "CommNet collapses" to "a modern CTDE baseline at competitive success collapses," which is the structural point.

I Reproducibility commands

An anonymized public repository accompanies this submission. The main paper artifacts were generated from the following scripts; `PYTHONPATH=.` and a Python 3 virtual environment with `numpy`, `scipy`, `pandas`, `matplotlib`, `imageio`, `gymnasium`, and `minigrid` are assumed throughout.

Computing infrastructure. All experiments ran on a single consumer machine: Apple Silicon (10 CPU cores used for the sweeps via `ProcessPoolExecutor`), macOS (Darwin 25.4), Python 3.9.6. Key library versions: `numpy` 1.x, `scipy`, `torch` (CommNet-PPO baseline and VMAS), `vmas` 1.5.2, `minigrid/gymnasium`. LLM experiments use local models via Ollama (`llama3.1:latest` 8B, `qwen3:4b`) at temperature 0 with deterministic on-disk response caching; the external-framework study additionally uses `openai` 2.44, `langgraph` 0.6, `langchain-ollama`, and `pyautogen` 0.9 (AutoGen classic). No GPU cluster or paid API was used; the full multi-agent block reproduces in $\approx$22 minutes on 8 cores, and each LLM study in minutes-to-hours of local inference.

One-command reproduction (multi-agent block, ≈ 22 minutes on 8 cores)

`bashscripts/reproduce_paper.sh`

Clean Paper 1 suite

- smoke suite (all local non-LLM blocks): `pythonscripts/run_paper1_clean_experiments.py--modesmoke--skip_llm--out_dirtmp/paper1_clean_experiments_smoke`
- full local suite (single-agent, oracle, multi-agent, CommNet, portability/noise): `pythonscripts/run_paper1_clean_experiments.py--modefull--skip_llm--workers8--out_dirtmp/paper1_clean_experiments_full`
- LLM TextNav step-budget sweep (requires local Ollama), run once per budget $K' \in \{6, 12, 25\}$: `python-mexperiments.llm_collective.run_model_sweep--modelsqwen3:4bllama3.1:latest--episodes8--step_limit<K'>--out_dirtmp/llm_steplimit_<K'>`

Single-agent (§3.2–§3.4)

- article benchmark and learned-baseline suite: `scripts/benchmark_symbolic_memory_article.py`
- article analysis and figure generation: `scripts/analyze_symbolic_memory_article.py`
- synthetic factorization sanity check: `scripts/validate_qrmc_factorization.py`
- oracle-stage intervention benchmark: `scripts/benchmark_oracle_stage_interventions.py`
- BabyAI transfer benchmark: `scripts/compare_semnav_babyai.py`

Multi-agent (§4.5)

- Base batch — peer at $K \in \{1, 2, 4, 8, 16\}$ *and* the three cadence-free architectures (independent, shared bus, central aggregator), 40 seeds (25,920 runs $=$ 16,200 peer $+$ 9,720 cadence-free), $\approx$ 18 min: `pythonexperiments/big_experiment/exp_cadence_phase.py--modefull--workers8--out_dirtmp/big_experiment_qrmc_40`
- Extended-cadence batch — peer at $K \in \{32, 48, 64\}$ only (9,720 runs), $\approx$ 4 min: `pythonexperiments/big_experiment/exp_extra_K.py--workers8--out_dirtmp/big_experiment_extraK`

  Note. These two batches partition the 35,640 total by *cadence range*; the architecture-wise partition of Appendix F.2 (25,920 peer over all eight cadences vs. 9,720 cadence-free) reuses the same two totals for *different* subsets (a coincidence of 3 extended cadences $\times$ 3,240 $=$ 3 cadence-free architectures $\times$ 3,240). The combined log is `tmp/paper1_clean_experiments_full/multiagent_cadence_full/runs.csv`.

- Q/R/M/C stage analysis and bootstrap CIs: `pythonexperiments/big_experiment/analyze_qrmc.py--runs_csvtmp/big_experiment_qrmc_40/runs.csv--out_dirtmp/big_experiment_qrmc_40`
- Multi-agent oracle interventions: `pythonexperiments/big_experiment/exp_oracle_interventions.py--workers8--out_dirtmp/big_experiment_oracle`
- Phase 1–4 smoke tests: `pythonscripts/run_all_smokes.py--out_dirtmp/all_smokes`
- PPO-trained CommNet baseline (Table 20, $\approx$ 96 min on a single CPU core): `python-mexperiments.commnet_ppo_baseline.train_ppo_commnet--total_updates200--rollouts_per_update64`
- Q/R/M/C eval of the PPO policy on 100 held-out seeds: `pythonexperiments/commnet_baseline/eval_with_qrmc.py--policy_pathexperiments/commnet_ppo_baseline/ppo_policy.pt--out_direxperiments/commnet_ppo_baseline--n_episodes100`
- MiniGrid-substrate portability sweep (§6, 100 runs, $\approx$ 2 min): `python-mexperiments.minigrid_multiagent_wrapper.run_portability_sweep--out_dirtmp/minigrid_portability`

J Asset and license note

The experiments use MiniGrid (Chevalier-Boisvert et al., 2023) and BabyAI (Chevalier-Boisvert et al., 2018), both released under the MIT License. The locally implemented baselines and benchmark runner scripts introduced in this paper will be released under the MIT License as part of the camera-ready artifact package.

Table 18: SOTA coverage for Paper 1. "Instrumented" means the run emits the operational $Q^\star, R^\star, M^\star, C^\star$ events used by Definition 1.

Family	Representative method	Status in this paper	Q/R/M/C comparison point
Topological navigation memory	SPTM-like graph baseline (Savinov et al., 2018)	Direct MiniGrid baseline	Tests whether topological memory without semantic query/lock exposes non-trivial stage events.
Behavior cloning	Learned grid-observation BC	Direct MiniGrid baseline	Strong success on easy tasks but no explicit query/lock interface, so diagnostics are structurally limited.
Learned communication	CommNet PPO (Sukhbaatar et al., 2016; Schulman et al., 2017)	Direct multi-agent baseline	Competitive success; $Q^\star$ saturates and R/M collapse, verifying observability loss.
Cooperative MARL (CTDE / value decomposition)	MAPPO (Yu et al., 2022) (trained); QMIX, MADDPG (Rashid et al., 2018; Lowe et al., 2017) (references)	**Direct multi-agent baseline**	A trained MAPPO (centralized critic) reaches 0.667 held-out success on the scarcity scenario and shows the same structural signature as CommNet: $Q^\star$ saturated at 1.00 and $P(M^\star \vert R^\star)$=1.00 (R/M collapsed) — observability loss is not an artifact of a weak baseline (Appendix H).
Reasoning/acting LLM agents	ReAct (Yao et al., 2023)	Primary-source reference	Q/R/M/C can instrument the resulting action traces if queries, retrieved evidence, target locks, and completions are logged.
Reflective LLM agents	Reflexion (Shinn et al., 2023)	Primary-source reference	Reflection can be treated as upstream Q/R revision; the framework asks whether it improves R, M, or C.
Embodied LLM agents	Voyager (Wang et al., 2023)	Primary-source reference	Skill/memory retrieval maps naturally onto R and M, but full reproduction is outside this paper.
Social/memory simulations	Generative Agents (Park et al., 2023)	Primary-source reference	Their observation/memory/reflection loop motivates logging stage-specific memory use rather than only believability.
Multi-agent LLM frameworks	AutoGen (Wu et al., 2023)	Primary-source reference	Conversation traces provide candidate Q/R/M/C events; this paper supplies the measurement contract.
Long-term LLM memory systems	MemGPT / Mem0 (Packer et al., 2023; Chhikara et al., 2025)	Primary-source reference	Persistent memory systems can be evaluated by where they fail: retrieval, materialization, or completion.
Local LLM substrate	qwen3:4b, llama3.1	Direct TextNav step-budget study	Q/R/M/C localizes a completion-limited failure ($C^\star \to 0$ while $Q/R/M$ stay saturated) in real LLM agents; see Table 4.

Table 19: Multi-agent oracle interventions on the scarcity cases at three cadences. Mean time-to-success (and success rate) averaged across $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$, two layouts (asymmetric, random), 15 seeds. Oracle R returns ground-truth water positions; oracle M locks the nearest real water; **oracle C** guarantees completion of a real materialised lock (Appendix F.14). Oracle C gives the largest *time* reduction (it removes travel) but the smallest *success* gain — completion is a time cost, not the success bottleneck, which sits upstream at R/M.

K	Baseline	Oracle R	Oracle M	Oracle C
mean t_{succ}				
1	10.33	7.22	7.21	**6.19**
4	10.98	7.22	7.21	**6.07**
16	8.45	7.22	7.21	**6.41**
success rate				
1	0.591	**0.614**	0.602	0.591
4	0.575	**0.614**	0.602	0.600
16	0.578	**0.614**	0.602	0.598

Metric	Train	Val	Test
Accuracy	0.584	0.553	0.590
Majority-class baseline accuracy	0.903	0.776	0.813
Precision @ recall $= 0.70$	0.171	0.278	0.275

Quantity	Value
$Q^\star$ rate	**1.00** (saturated)
$R^\star$ rate	0.50
$M^\star$ rate	0.50 (collapses onto R)
$C^\star$ rate	0.19
$P(R \mid Q)$	0.50
$P(M \mid R)$	1.00
$P(C \mid M)$	0.42
Success rate	0.19
Mean steps	80 (budget exhausted)

Table 20: REINFORCE vs PPO CommNet vs symbolic peer (K=8) on the scarcity scenario (N=3, T=2, asymmetric, hazard 0.05). PPO is competitive on success rate, yet still exhibits Q-saturation and R/M collapse — the diagnostic claim verified, not asserted.

Metric	CommNet REINFORCE	**CommNet PPO**	Symbolic peer (K=8)
Success rate	0.19	**0.667**	0.65
$Q^\star$ rate	1.00	**1.00**	0.99
$R^\star$ rate	0.50	**0.997**	0.69
$M^\star$ rate	0.50	**0.997**	0.69
$P(M^\star \mid R^\star)$	1.00	**1.00**	0.69
$P(C^\star \mid M^\star)$	0.42	**0.67**	0.87
R/M separated?	no	**no**	**yes**

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Justification: The paper describes a unified benchmark runner and source-of-truth artifacts. An anonymized public release of the codebase — single-agent benchmark, multi-agent grid environment, 35,640-run sweep driver, Q/R/M/C analyser, oracle-intervention scripts, and exact commands to reproduce every table and figure — accompanies the submission as supplemental material (anonymized GitHub repository; single-command reproduction of the main multi-agent sweep on an 8-core machine in $\approx$ 22 minutes, matching Appendix H).

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Answer: [Yes]

Justification: The benchmark protocol, task definitions, assist settings, seeds, episode counts, baseline families, and the learned baselines are described in the main text together with the measurement and statistical protocols: single-agent setup in Section 3.1, multi-agent sweep in Section 4.4, and per-baseline hyperparameters in Appendix G.

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